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Inherited Games Studio (IGS)

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Dedication

To my beloved family,

who have been my unwavering pillar of strength
throughout every step of this journey.

To my parents,
whose sacrifices, endless encouragement, and unconditional love
have been the foundation upon which I built every dream.

To my brother Oussema and my sister Emna,
for the laughter, the support, and the belief
that always pushed me forward.

To my friends and colleagues,
who made these long study years memorable
and filled them with moments of joy and solidarity.

And to myself —
for the late nights, the persistence, and the courage
to see this through to the end.

This work is for all of you.

Asma Daoussi

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Asma Daoussi
Sousse, 2026

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List of Acronyms

Acronym	Definition
API	Application Programming Interface
CDN	Content Delivery Network
CORS	Cross-Origin Resource Sharing
CRUD	Create, Read, Update, Delete
DB	Database
DM	Direct Message
ERP	Enterprise Resource Planning
GLB	GL Binary (3D format)
GLTF	GL Transmission Format
HDRI	High Dynamic Range Imaging
HTTP	HyperText Transfer Protocol
IGS	Inherited Games Studio
ISITCOM	Institut Supérieur de l'Informatique et des Technologies de Communication
JWT	JSON Web Token
MVC	Model-View-Controller
OAuth	Open Authorization
ORM	Object-Relational Mapping
PBR	Physically Based Rendering
PFE	Projet de Fin d'Études (End of Studies Project)
RBAC	Role-Based Access Control
REST	Representational State Transfer
SPA	Single Page Application
SQL	Structured Query Language
SSO	Single Sign-On
UML	Unified Modeling Language
UUID	Universally Unique Identifier

General Introduction

The global video game industry has experienced remarkable growth over the past decade, evolving from a niche entertainment segment into one of the most lucrative creative economies in the world. With revenues surpassing hundreds of billions of dollars annually and a continuously expanding ecosystem of developers, studios, and independent creators, the demand for high-quality digital assets has never been greater. Beyond the games themselves, a significant and often overlooked opportunity lies not in the creation of video games directly, but in building the infrastructure that supports them, namely, digital marketplaces dedicated to the exchange of 3D assets. These platforms serve as essential bridges between creators and consumers, enabling studios, developers, architects, and designers to access a vast library of ready-made models, textures, animations, and environments. The rise of real-time 3D technologies, game engines such as Unity and Unreal Engine, and the expanding metaverse economy has further accelerated the need for structured, accessible, and reliable 3D asset marketplace solutions. Investing in such platforms represents a strategic and high-value proposition in today's digital economy, as they reduce production costs, shorten development cycles, and democratize access to professional-grade visual content.

For this purpose, this project introduces a unified 3D digital marketplace — a centralized platform designed to connect creators and consumers of 3D assets within a single ecosystem. The platform is built as a two-sided system: a public-facing **IGS Marketplace** where external users can browse, upload, purchase, and sell assets across multiple categories, and a private **IGS Workspace** reserved for internal company staff, covering project management, team collaboration, content moderation, and administrative control. By combining both sides under one shared backend, the solution addresses the fragmentation of the 3D asset market while also providing IGS with the internal tooling it needs to operate the platform efficiently.

This report is structured as follows:

- **Chapter 1** presents the host organization, the project context, a comparative study of existing solutions, and the chosen Scrum methodology.
- **Chapter 2** defines the system actors, functional and non-functional requirements, product backlog, and logical architecture.
- **Chapter 3 — Marketplace Foundations** covers Sprints 1 and 2: authentication, RBAC, the public asset catalog, 3D and WebXR previews, cart, payment,

and download library.

- **Chapter 4 — Seller Workflow and Workspace** covers Sprints 3 and 4: asset upload, moderation, reviews, Creator Studio, and the full internal IGS Workspace.
- **Chapter 5 — Advanced Features** covers Sprint 5: the AI chatbot, sentiment analysis, recommendations, security hardening, WebXR immersive preview, and the planned deployment.

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General Context

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Introduction

This opening chapter establishes the full context of the project by introducing the host organization and the motivations behind the platform initiative. It begins by presenting Inherited Games Studio, its mission and technical positioning, before examining the problem statement that led to the creation of the IGS Marketplace. A comparative study of existing digital asset marketplaces is then conducted, analyzing their strengths and limitations, and leading to the articulation of the proposed solution. The chapter concludes by detailing the Agile Scrum development framework — explaining the team roles, sprint structure, and overall project planning timeline.

1.1 Presentation of the Host Organization

Inherited Games Studio (IGS) is a game development studio that creates educational and interactive games using immersive technologies and AI-driven mechanics.

A. Mission

IGS develops games that combine learning with entertainment, using VR, AR and AI to create engaging experiences for users.

Figure 1.1: Inherited Games Studio Logo

B. Methodology

IGS employs several key strategies: immersive VR/AR environments for spatial learning, AI systems that adapt gameplay to player behavior, multiplayer and gamification for engagement, data-driven refinement, and balanced edutainment design.

1.2 Project Presentation

This section examines the market landscape that motivated the project and the specific problems it was built to address.

1.2.1 Context and Motivation

The global digital-content economy has entered a phase of sustained growth. Recent market research [1] values the worldwide **3D model marketplace** at approximately **USD 1.92 billion in 2023**, with a forecast of **USD 4.86 billion by 2031** and a compound annual growth rate of **11.2%**. A complementary study [2] attributes most of this demand to three industry verticals: video games (**38.2%** of consumption), architectural visualization (**22.1%**) and film and television production (**17.8%**).

This global growth directly shaped the challenges encountered by IGS, as detailed in the following subsection.

1.2.2 Problem Statement

Despite this global trajectory, Tunisia did not have any dedicated 3D digital asset marketplace. Local developers and studios were forced to rely on foreign platforms.

Beyond the market gap, IGS itself lacked a centralized internal system to manage its own assets, coordinate employee activities, supervise trainees and monitor platform operations. Critical tasks were handled through disconnected tools, creating friction and reducing traceability across the organization.

The project addresses two intertwined problems. The first is external: the absence of a national platform through which Tunisian creators can publish, price and sell three-dimensional content while receiving payment in local currency. The second is internal: the absence within IGS of an integrated management environment that governs asset workflows, access rights, team collaboration and operational visibility under a single, coherent system.

To better understand what the proposed solution should offer, a survey of the leading existing platforms was conducted.

1.3 Comparative Study of Existing Solutions

The following subsections examine each major platform individually and identify the gaps that justified building a dedicated solution.

1.3.1 Analysis

The international landscape is dominated by several major platforms.

Each major platform is examined in detail below.

1.3.2 Platform Comparison

Unity Asset Store

Launched in 2010, Unity Asset Store is tightly integrated with the Unity Editor, offering millions of assets. However, it is engine-specific, lacks RBAC, and provides no internal team tools.

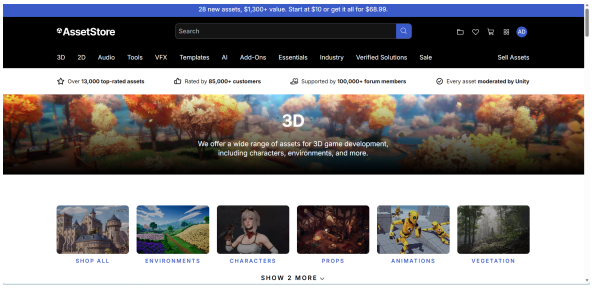

Figure 1.2: Unity Asset Store [3]

Fab (Epic Games)

Fab, Epic Games' unified marketplace, supports multiple engines and benefits from Unreal Engine integration. Despite its breadth, it lacks customization, internal workflows, and AI-powered features.

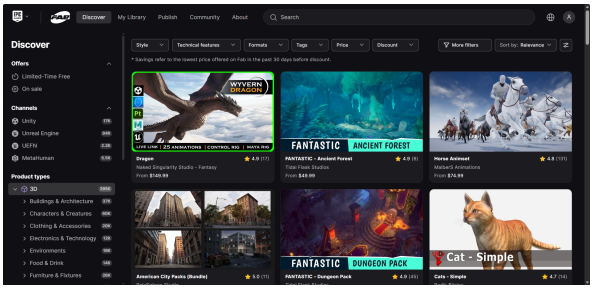

Figure 1.3: Fab Marketplace [4]

Sketchfab

Sketchfab excels in in-browser 3D visualization and supports multiple formats. It is widely used for showcasing models but does not provide monetization flexibility, RBAC, or internal studio management.

CGTrader

CGTrader offers a large catalog and supports multiple engines. However, it is externally hosted, lacks internal collaboration tools, and does not integrate AI or role-based workflows.

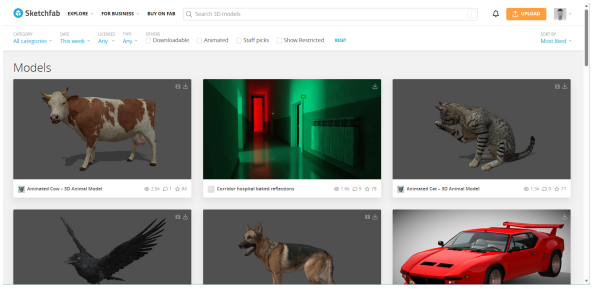

Figure 1.4: Sketchfab Platform [5]

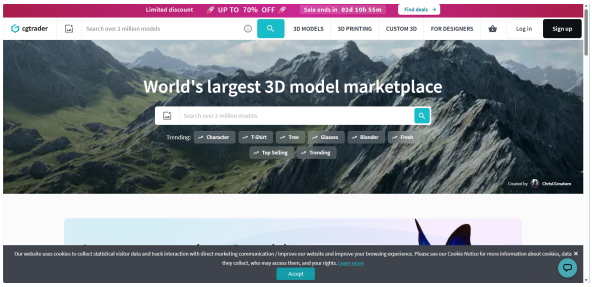

Figure 1.5: CGTrader Marketplace [6]

The key observations from this analysis are synthesized in the comparison table below.

1.3.3 Comparison table of platforms

Table 1.1: Comparison of existing 3D asset marketplace platforms

Feature	Unity	Fab	Sketchfab	CGTrader	IGS Marketplace
Multi-engine support	×	✓	✓	✓	✓
In-browser 3D preview	Limited	Limited	✓	✓	✓
WebXR / VR-AR preview	×	×	×	×	✓
Custom RBAC	×	×	×	×	✓
Internal team tools	×	×	×	×	✓
AI recommendation	×	×	×	×	✓
AI chatbot	×	×	×	×	✓
Intern access control	×	×	×	×	✓
Local payment gateway	×	×	×	×	✓
Open/self-hosted	×	×	×	×	✓

The gaps identified across all reviewed platforms directly informed the design of the proposed solution.

1.3.4 Proposed Solution

The project developed for IGS consists in a unified digital platform dedicated to the exchange and management of three-dimensional creative assets. It was built as a **two-sided system**.

The first side, called **IGS Marketplace**, is a public-facing storefront open to visitors, buyers and independent creators. It allows external sellers to publish and monetize their work and gives buyers a structured catalog covering thirteen asset categories, with secure transactions and a personal download library. A distinctive feature of this storefront is its **immersive preview capability**: beyond a conventional thumbnail, visitors can manipulate a three-dimensional asset interactively inside the browser and step into the asset in **virtual reality / augmented reality**.

The second side, called **IGS Workspace**, is a private dashboard reserved for company collaborators. It supports the internal life of the studio through project management, team communication, supervised onboarding of trainees, content moderation and access to a full suite of administrative tools. It sits at the intersection of an **ERP** — governing internal processes and resources — and a **CRM** — managing relationships and communication with the platform's user community.

Although the two interfaces serve different audiences, they share a common foundation that unifies user identity, access-control logic and the data layer. The platform additionally integrates intelligent components — a conversational assistant, an automated review analyser and a personalized recommendation engine — that enrich the user experience without altering the core business workflows.

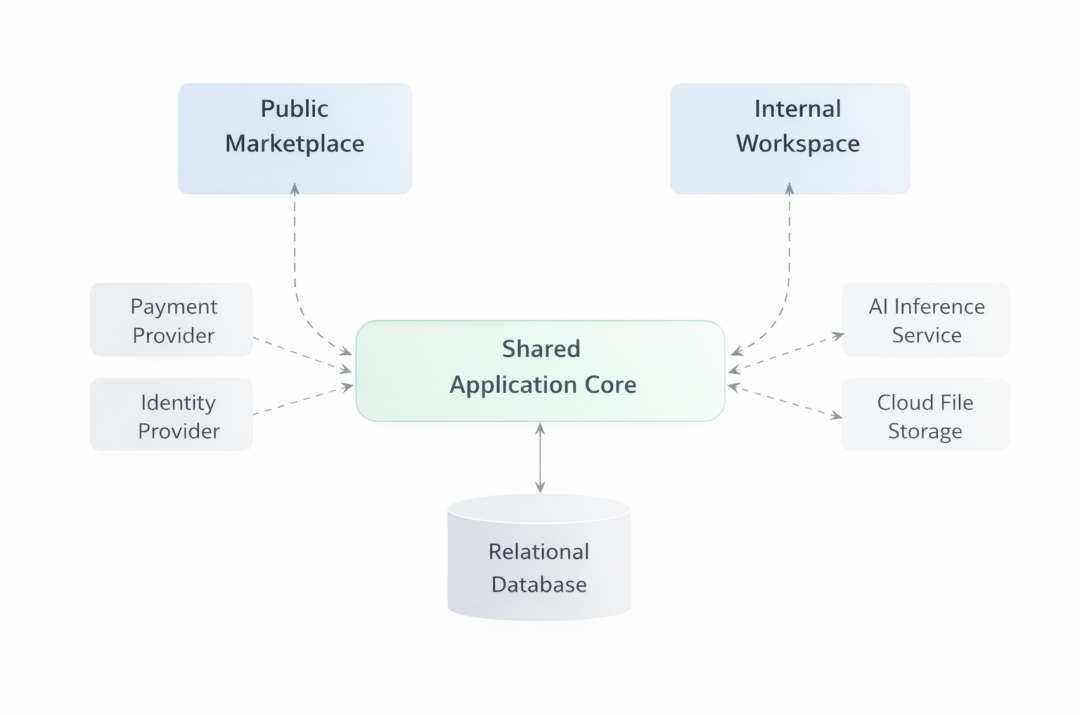

Figure 1.6: Conceptual architecture of the proposed solution

With the solution defined, the next step was to select a development methodology suited to the project's scope and constraints.

1.4 Work Methodology

Three widely used approaches were examined before selecting the one most appropriate for this internship context.

1.4.1 Comparative Study of Methodologies

Before selecting a development approach, three widely adopted software engineering methodologies were examined: the Waterfall model, the V-Cycle model, and the Agile Scrum framework. Each carries distinct assumptions about requirements stability, feedback cycles, and risk management.

A. Waterfall Model

The Waterfall model is a **sequential, plan-driven** approach in which each phase (requirements, design, implementation, testing, deployment) must be completed before the next one begins. All requirements are frozen at the outset, making it poorly suited to projects where scope evolves during development.

B. V-Cycle Model

The V-Cycle extends Waterfall by pairing each development phase with a corresponding testing phase. Testing is planned early, improving defect traceability. However, it inherits the same rigidity as Waterfall and delivers a working product only at the end.

C. Agile Scrum Framework

Scrum is an **iterative and incremental** framework built on short development cycles called **sprints** (typically two to four weeks). At the end of each sprint a shippable product increment is delivered and reviewed by stakeholders. Three core roles structure the team: *Product Owner*, *Scrum Master*, and *Development Team*.

D. Side-by-Side Comparison

Table 1.2: Comparison of development methodologies

Criterion	Waterfall	V-Cycle	Agile Scrum
Requirements	Fixed upfront	Fixed upfront	Evolving backlog
Delivery	Single final release	Single final release	Incremental sprints
Stakeholder feedback	At delivery only	At delivery only	Every sprint
Testing integration	End of project	Paired with each phase	Continuous (per sprint)
Flexibility to change	Very low	Low	High
Risk management	Late detection	Earlier via test plans	Continuous mitigation
Best suited for	Stable, well-defined	Safety-critical systems	Evolving, collaborative

Based on this comparison, the Agile Scrum framework was selected for its adaptability and incremental delivery model.

1.4.2 Methodology Adopted: Scrum

Scrum was selected for this internship project because it supports adaptive planning, incremental delivery, and regular validation. The application requirements evolved

during the internship, and the supervisors were able to review deliverables at the end of each sprint.

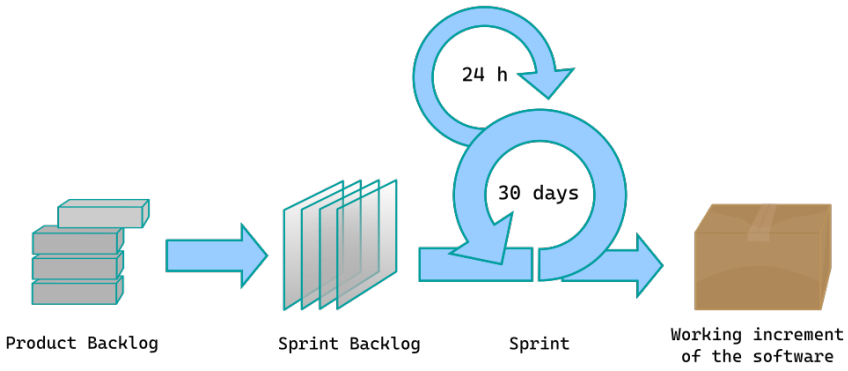

Figure 1.7: Scrum process overview [7]

The effective application of Scrum relies on three distinct roles, each carrying specific responsibilities within the team.

1.4.3 Scrum Process Roles

The Scrum framework defines clear roles that were adopted for this project:

- **Product Owner**: Owns the product vision, sets priorities, and manages the backlog.
- **Scrum Master**: Facilitates the process, removes impediments, and ensures Scrum practices are followed.
- **Development Team**: Builds the product increment, implements features, and delivers functionality.

Figure 1.8: Scrum Team Roles

With the roles clearly defined, the following subsection describes how the Scrum process was concretely applied throughout the project.

1.4.4 Implementation of the Scrum Methodology

The Scrum methodology was implemented using sprint planning, backlog refinement, and sprint reviews. Tasks were tracked and prioritized using ClickUp.

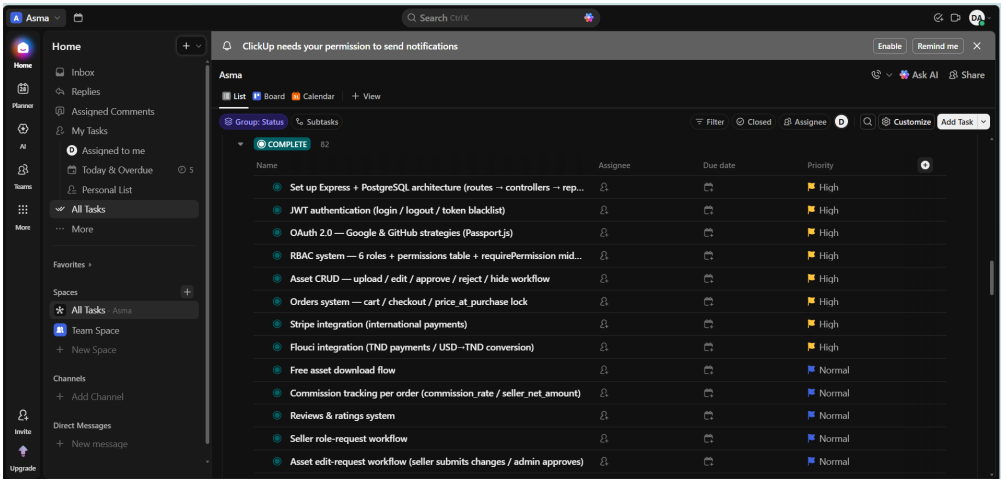

Figure 1.9: ClickUp task board

The chosen methodology directly shaped the project planning, as detailed in the following section.

1.5 Planning

Project planning was driven by sprint sequencing, milestone delivery, and scope tracking. The development timeline spans eighteen weeks across five sprints, with each sprint delivering a coherent slice of functionality.

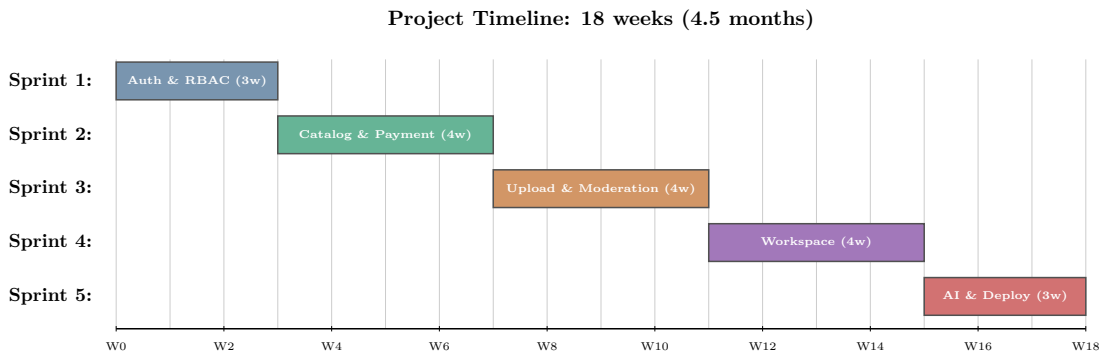

Figure 1.10: Project Timeline — 18-week Gantt chart across five Scrum sprints

Conclusion

This chapter introduced the host organization, framed the project's motivations, and analyzed the competitive landscape of existing solutions before presenting the proposed IGS Marketplace platform. It also established the Scrum-based development methodology and the sprint timeline that will guide the work throughout this report. With this context in place, the next chapter defines the system actors, functional requirements, and overall architecture.

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Introduction

This chapter defines the system scope and lays the conceptual foundation for the entire development effort. It opens by identifying the actors who interact with the platform, distinguishing external marketplace users from internal IGS collaborators under a role-based access control model. The functional and non-functional requirements are then formalized and organized into a product backlog distributed across five sprints. To visualize the system's behavior and structure, the chapter presents global use case and class diagrams for both the Marketplace and the Workspace. It closes by describing the three-tier architecture, the design patterns applied, and the hardware, software, and cloud environment used during development.

2.1 Actors Identification

The platform is used by external marketplace users and by internal IGS collaborators. Access rights are controlled through a role-based access control model (RBAC), where each role receives a set of permissions adapted to its responsibilities.

— **Member**

— **Seller**

— **Intern**

— **Employee**

— **Administrator**

— **Super Administrator**

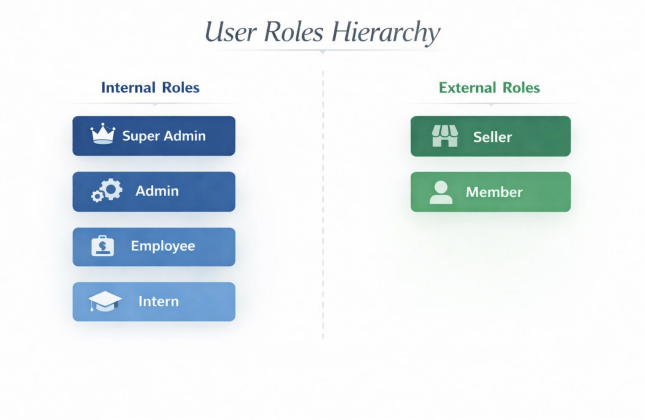


Figure 2.1: User Role Hierarchy

With the actors identified, their respective functional and non-functional requirements can now be specified.

2.2 Needs Identification

The functional requirements describe what each category of actor is expected to accomplish through the platform.

2.2.1 Functional Requirements

Visitor:

— **Register an Account:** create an account using email/password or through an OAuth provider (Google, GitHub).

— **Browse the Catalog:** browse the public catalog by type(models, textures, shaders ...), search by name, and filter.

— **Preview an Asset:** inspect asset details and preview them in the browser or in VR/AR.

— **Use the AI Chatbot:** chat with a bot.

— **Get Recommendations:** consult trending asset suggestions.

Member :

— **Authenticate:** sign in, sign out and recover the account through a password-reset email.

— **Manage Profile:** view and update personal information (display name, bio, avatar, password).

— **View Notifications:** consult the notification feed for orders, requests and moderation events.

— **Manage Cart:** add and remove assets from the personal cart.

— **Checkout and Pay:** pay through Stripe.

— **Manage Library:** access and download free or purchased assets.

— **Manage Reviews:** rate, write, edit and like reviews on owned assets.

— **Request Seller Access:** submit a justified request to be promoted to the seller role.

— **Use the AI Chatbot:** ask marketplace questions in natural language with personalized context.

— **Get Recommendations:** consult personalized asset suggestions based on purchase and download history.

Seller:

— **Upload an Asset:** publish assets with files, thumbnails, metadata and pricing.

— **Request Asset Edit:** request modifications on already-approved assets without bypassing moderation.

— **Access Creator Studio:** consult personal sales statistics and per-asset performance.

— **Open Support Tickets:** communicate with the moderation team through a ticketing system.

— **Manage Trades:** propose, accept, reject or cancel asset exchanges.

Intern (Workspace + Marketplace):

— **Request Asset Access:** request access to paid marketplace assets with a written justification.

— **Redeem Vouchers:** redeem discount codes issued by authorized staff.

— **Manage Tasks:** create, assign, update and track tasks on a Kanban board.

— **Participate in Projects:** consult assigned projects.

— **Use Team Chat:** communicate in real time via channels and direct messages.

— **Use the Calendar:** consult task deadlines, project milestones and personal events.

— **Manage Calendar Events:** create, edit and delete personal events on the calendar.

Employee:

— **Approve Intern Asset Requests:** approve or reject intern asset access requests with a note.

— **View Reports:** consult platform analytics on revenue, sales and per-category performance.

— **Track Performance:** consult personal and team performance metrics.

— **Create Projects:** create projects, assign members and monitor progress.

Administrator:

— **Manage Users:** create internal users, assign or revoke roles and update account status.

— **Manage Role Requests:** approve or reject member-to-seller upgrade requests.

— **Moderate Assets:** approve, reject or hide submitted catalog assets and intern uploads.

— **Manage Categories:** create, edit and deactivate asset categories.

— **Manage Vouchers:** issue, edit and revoke voucher codes.

— **Manage Orders:** consult orders and update their status.

— **Manage Teams:** create teams, add or remove members and assign team roles.

— **Handle Support Tickets:** consult seller tickets, reply and update ticket status.

Super Administrator:

— **Consult Audit Logs:** consult the full audit trail of all platform actions with filtering.

— **Configure Roles & Permissions:** consult and edit the permission set attached to each role.

— **Configure Platform Settings:** control global configuration including the commission rate.

Alongside these functional capabilities, the system must also satisfy a set of quality and reliability constraints.

2.2.2 Non-Functional Requirements

Security : the application must guarantee the confidentiality and integrity of user data. Access must be controlled based on roles and permissions, and all sensitive actions must be traceable.

Performance : the application must maintain a minimal response time and remain responsive even with large volumes of content and simultaneous users.

Reliability : the application must always be able to function correctly. Errors must be handled gracefully without disrupting the user experience.

Scalability : the system must be designed to support growth in users, assets and traffic without requiring major structural changes.

Usability : the system must be simple and intuitive to use. Navigation between pages must be smooth, and the interface must be accessible across different devices.

Maintainability : the architecture must make it easy to add new features, fix issues and apply updates without impacting the rest of the system.

These requirements were organized into a prioritized backlog and distributed across five development sprints.

2.3 Project Management with Scrum

The project scope is organized as a product backlog. Each backlog item represents a user story or technical capability, prioritized by importance and assigned to one of five sprints.

The backlog below lists every user story alongside its assigned sprint and priority level.

2.3.1 Product Backlog

Table 2.1: Product Backlog — IGS Marketplace

ID	Feature	User Story	Sprint	Priority
1	Authentication	As a user, I can register, sign in, sign out and recover my account securely.	S1	**High**
2	RBAC	As an administrator, I can assign roles and permissions so each user sees only what they are allowed to.	S1	**High**
3	User Profile	As a user, I can view and update my personal profile information.	S1	**High**
4	Catalog	As a visitor, I can browse and filter the public asset catalog by category and keyword.	S2	**High**
5	Search & Filters	As a visitor, I can narrow results by category, price, rating, file type and availability.	S2	**High**
6	Asset Preview	As a buyer, I can inspect assets in 3D, VR and AR before purchasing.	S2	**High**
7	Cart	As a member, I can add, remove and review cart items before checkout.	S2	**High**
8	Payment & Orders	As a buyer, I can pay securely and receive a confirmed order record.	S2	**High**

ID	Feature	User Story	Sprint	Priority
9	Library & Downloads	As a buyer, I can access and download all assets I have unlocked.	S2	**High**
10	Seller Request	As a member, I can request seller access by providing a justification.	S3	**Medium**
11	Asset Upload	As a seller, I can upload files, thumbnails, metadata, tags, formats and pricing.	S3	**High**
12	Moderation	As an admin, I can approve, reject, hide or request changes on submitted assets.	S3	**High**
13	Asset Edit Request	As a seller, I can request modifications on approved assets without bypassing moderation.	S3	**Medium**
14	Reviews & Ratings	As a buyer, I can rate and review assets I own.	S3	**Medium**
15	Notifications	As a user, I can receive notifications about requests, orders and moderation events.	S3	**Medium**
16	Trades	As a user, I can propose and respond to peer-to-peer asset trades.	S3	**Low**
17	Intern Requests	As an intern, I can request access to paid assets for production needs.	S4	**High**
18	Vouchers	As an admin, I can issue vouchers; as a user, I can redeem them.	S4	**Medium**
19	Administration	As an admin, I can manage users, roles, assets, categories, orders and settings.	S4	**High**
20	Analytics & Audit	As an admin, I can consult analytics and trace actions through audit logs.	S4	**Medium**
21	Workspace Projects	As internal staff, I can create projects, assign members and monitor progress.	S4	**High**
22	Workspace Tasks	As internal staff, I can create, assign, update and track tasks.	S4	**High**
23	Workspace Chat	As internal staff, I can communicate in real time via channels and direct messages.	S4	**Medium**
24	AI Review Analysis	As an admin, I can use sentiment labels to support moderation decisions.	S5	**Medium**
25	Recommendations	As a user, I can receive personalized asset recommendations.	S5	**Medium**
26	Chatbot	As any user, I can ask marketplace questions through the AI assistant.	S5	**Medium**

To complement the textual backlog, global use case diagrams provide a visual overview of the interactions between actors and the system.

2.3.2 Global Use Case Diagrams

The project is composed of two applications sharing the same backend: the public **IGS Marketplace** and the internal **IGS Workspace**. Each one has a distinct set of actors and capabilities, so a dedicated use case diagram is presented for each.

Marketplace — Use Case Diagram

The figure below presents an overview of the functional behavior of the Marketplace. This diagram represents the interactions between the actors and the use cases of the system.

Figure 2.2: Use Case Diagram — IGS Marketplace

Workspace — Use Case Diagram

The Workspace is the internal managemeent layer reserved for company staff.

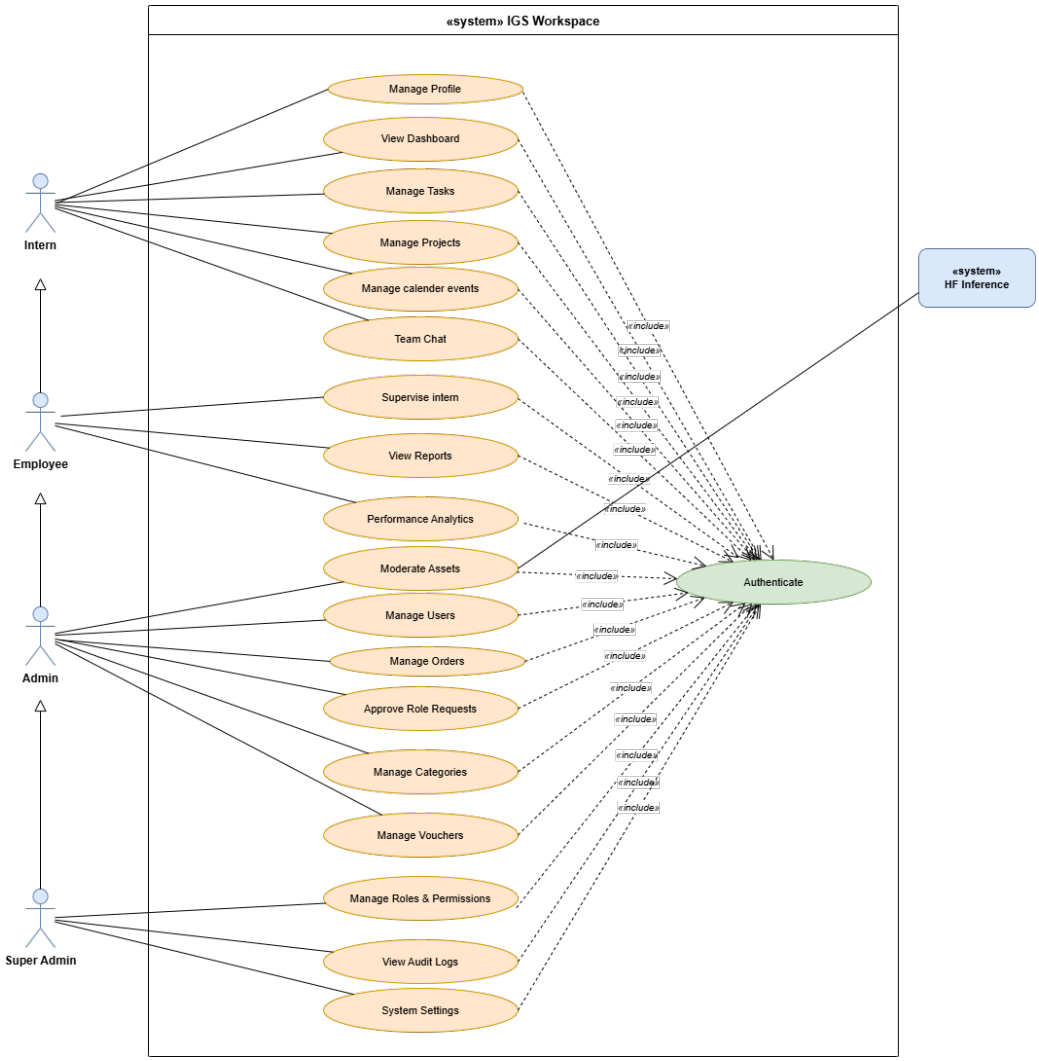


Figure 2.3: Use Case Diagram — IGS Workspace

The structural view of the system is completed by class diagrams that model the key entities and their relationships.

2.3.3 Global Class Diagrams

Two global class diagrams are presented, one per application. They are derived from the PostgreSQL schema and the backend repositories.

The first diagram covers the public marketplace domain.

2.3.4 Marketplace — Class Diagram

The Marketplace domain centres on the `User` entity (acting as both buyer and seller), connected to `Asset`, `Order`, `OrderItem`, `CartItem`, `Review`, `Trade` and `InternAssetRequest`.

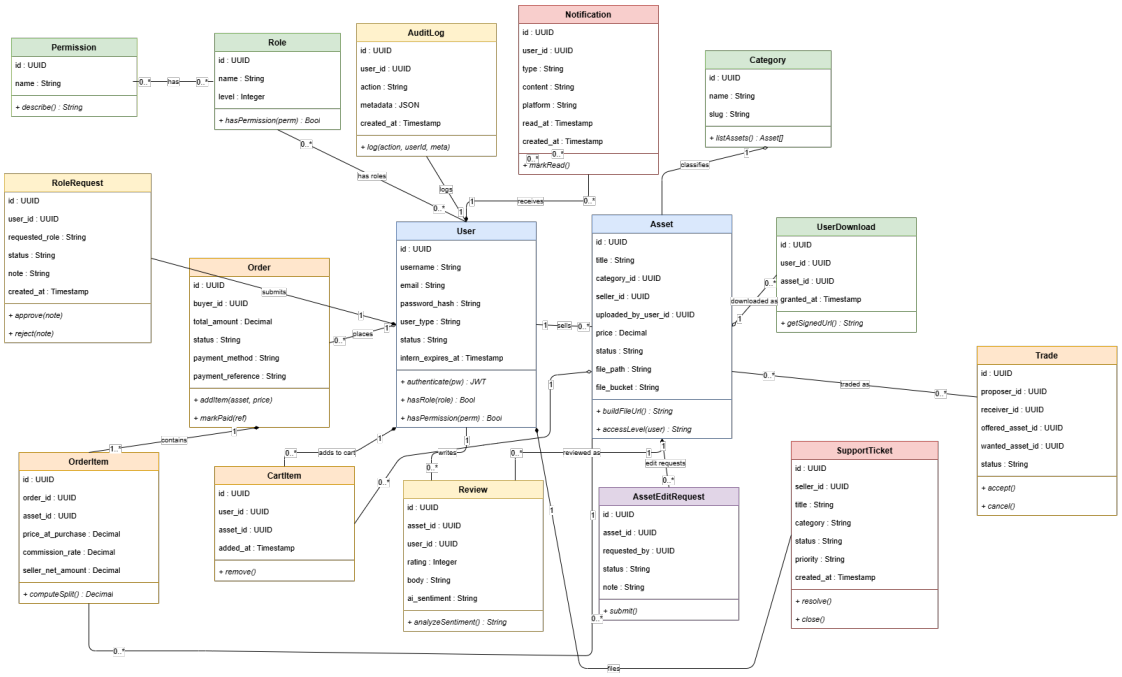


Figure 2.4: Class Diagram — IGS Marketplace

The second diagram covers the internal workspace domain.

Workspace — Class Diagram

The Workspace domain reuses the same `User` entity but adds the RBAC model (`Role`, `Permission`), the project management entities (`Project`, `ProjectMember`, `WorkspaceTask`), the real-time chat (`ChatChannel`, `Message`), the `Notification` feed and the `AuditLog` trail.

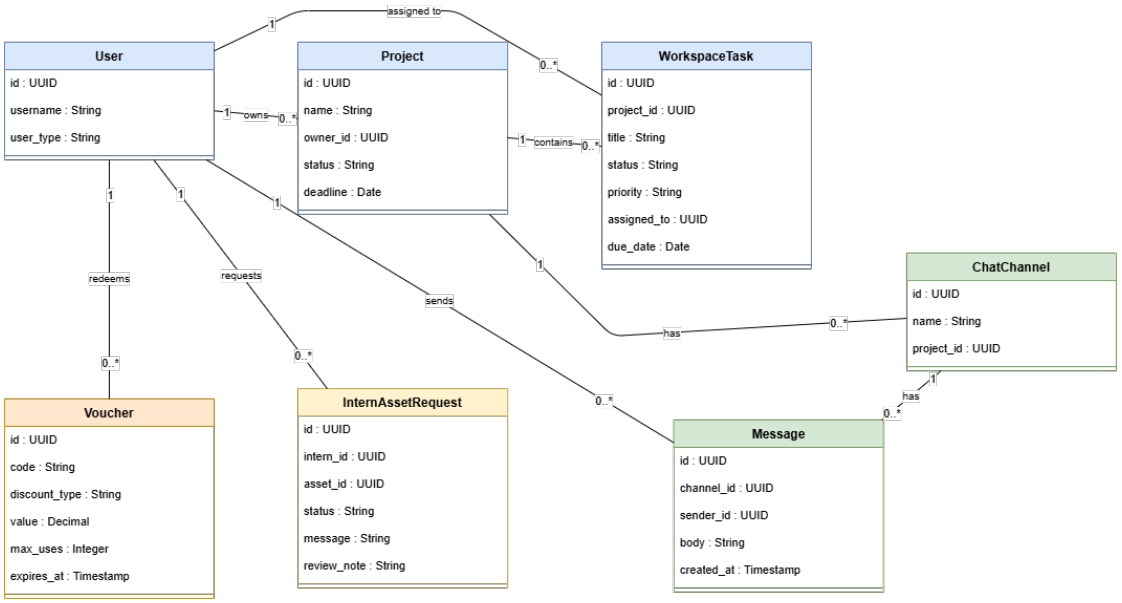


Figure 2.5: Class Diagram — IGS Workspace

Having established the domain model, the overall system architecture is presented in the following section.

2.4 Project Architecture: 3-Tier

The project follows a **three-tier architecture**:
Three-Tier Architecture, also known as three-layer architecture, organizes applications into three logical and physical layers: the presentation layer, the business logic layer, and the data access layer. This separation allows each layer to be developed, maintained, and scaled independently, improving flexibility and reducing the risk of system-wide issues when changes are made.

- **Presentation Layer (UI Layer):** Handles the user interface and interaction, displaying data and capturing user input.
- **Business Logic Layer (Application Layer):** The core of the application where rules, workflows, and data processing occur.
- **Data Access Layer (Persistence Layer):** Manages storage, retrieval, and management of data, abstracting databases and external APIs from upper layers.

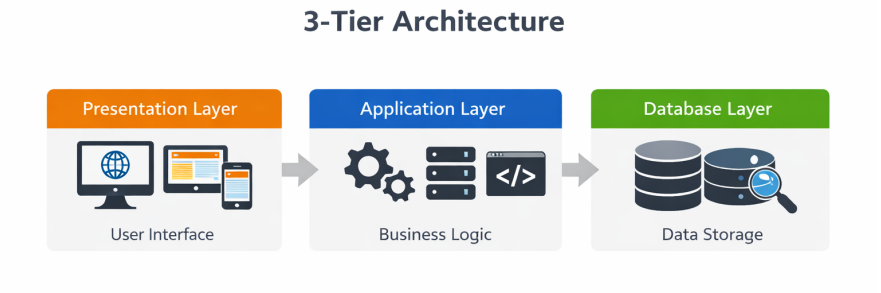


Figure 2.6: Three-Tier Architecture

2.4.1 Backend Layered Architecture

The backend maps the business-logic tier onto four internal layers:

- **Routes layer** (`routes/`) — declares HTTP endpoints, chains middleware (JWT authentication, role/permission checks, rate limiters, input validators) and delegates execution to a controller.
- **Controllers layer** (`controllers/`) — handles the HTTP request and response objects, orchestrates calls to repositories and services, and formats the JSON reply.

- **Repositories layer** (`repositories/`) — contains all raw parameterised SQL executed through the PostgreSQL connection pool; no SQL appears outside this layer.
- **Services layer** (`services/`) — encapsulates cross-cutting concerns: file storage (Supabase), audit logging, token revocation, AI inference calls and recommendation scoring.

This strict separation means that swapping the database engine, replacing a cloud service or adding a new route never requires changes to more than one layer.

2.4.2 Frontend Architecture

Both the Marketplace and the Workspace are built as independent React single-page applications using **HashRouter** (paths prefixed with `#/`), so the web server never needs to handle deep-link URLs — only the root path is served.

Application-wide state is managed through a chain of React Context providers mounted at the root of each app:

1. `AuthContext` — stores the JWT and the decoded user profile; handles OAuth callback parsing and logout.
2. `PermissionsContext` — exposes helper functions `can(permission)`, `hasRole(role)` and `hasMinLevel(n)` used by every protected route and conditional UI element.
3. `CartContext` — mirrors the server-side cart and synchronises additions and removals in real time.
4. `NotificationContext` — subscribes to Socket.IO events and maintains the unread notification counter across pages.

All HTTP calls are routed through a shared **Axios instance** (`services/api.js`) that automatically attaches the `Authorization: Bearer` header from local storage, so individual pages never handle authentication headers manually.

In addition to the three-tier structure, several design patterns were applied at the code level to ensure modularity and testability.

2.5 Other used Design Patterns

Five core design patterns were applied across the codebase for modularity, testability and maintainability:

Pattern	Purpose
Repository Pattern	Encapsulates all SQL queries in dedicated repository modules; controllers never write SQL directly.
Singleton	Ensures a single shared instance of critical services (database pool, token blacklist, storage client) is reused across the application.
Provider Pattern	Shares state globally across React components without prop drilling; context subscribers receive updates automatically.

The work environment that supported the development process is described in the following section.

2.6 Work Environment

Development took place on dedicated hardware and relied on a curated set of tools covering the full delivery chain.

2.6.1 Hardware Environment

The application was developed and tested on a personal laptop with the following specifications:

- **Processor:** Intel Core i7 (8^{th} generation)
- **RAM:** 16 GB DDR4
- **Storage:** 512 GB SSD
- **Operating System:** Windows 11 Pro

For the virtual and augmented reality testing, the following headset was used:

— **Meta Quest 3** [8] A standalone mixed-reality headset manufactured by Meta Platforms, featuring full-colour passthrough cameras for augmented reality and a high-resolution display for virtual reality. It was used throughout the project to validate the WebXR immersive preview feature of the marketplace directly on the target hardware.

The hardware configuration was supported by a set of software tools that streamlined development and version control.

2.6.2 Software Environment

The development relied on a carefully selected set of open-source and cloud-based tools spanning the entire delivery chain: code editing, version control, automated integration, database management, design and documentation.

— **Visual Studio Code** [9] A lightweight, open-source source-code editor developed by Microsoft, available on Windows, macOS and Linux. It was the primary development environment for both the frontend and backend codebases, extended with ESLint, Prettier and REST-client plugins.

— **GitHub** [10] A cloud-based platform for Git version control and collaborative software development. The entire project source code was hosted on a private GitHub repository, enabling branch-based development, pull requests and automated CI/CD pipelines.

— **Hostinger** [11] A cloud hosting provider offering Virtual Private Server infrastructure. The backend API and the built frontend assets were deployed on a Hostinger VPS managed by Nginx as a reverse proxy and PM2 as a process supervisor.

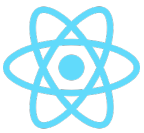
— **draw.io** [12] A free, browser-based diagramming tool used to produce the UML diagrams, architecture views and flowcharts included in this report.

— **ESLint** [13] A static analysis tool for JavaScript that enforces code-style rules and catches common errors before runtime. It was configured with a shared rule-set across both frontends and the backend.

The application itself was built on a carefully selected stack of open-source frameworks and libraries.

2.6.3 Frameworks and Libraries

— **React** [14] A declarative, component-based JavaScript library for building user interfaces, developed and maintained by Meta. Both the public marketplace and the internal workspace were built as independent React single-page applications.



— **Socket.IO** [15] A bidirectional, event-based communication library built on top of WebSockets. It was used to implement the real-time team chat and the live notification feed shared between the marketplace and the workspace.

— **Express.js** [16] A minimal and flexible Node.js web framework chosen to expose the REST API and compose the security middleware chain covering authentication, authorization, rate limiting and input validation.

Data storage and cloud infrastructure were delegated to specialized external services, as described below.

2.6.4 Data Persistence and Cloud Services

— **Supabase** [17, 18] An open-source Backend-as-a-Service platform built on PostgreSQL. It was used for cloud file storage: a public bucket for thumbnails and profile pictures, and a private bucket for three-dimensional asset files served through time-limited signed URLs.

— **Stripe** [19] An international payment processing platform used to handle card payments through the Payment Intents API. It was integrated alongside a local Tunisian gateway to support both international and dinar-denominated transactions.

— **Hugging Face** [20] A machine learning platform providing hosted inference APIs. It was used to run the DistilBERT sentiment classification model for automatic review analysis and the Qwen2.5-72B-Instruct large language model that powers the AI chatbot.

Conclusion

This chapter established the full scope of the system by identifying actors, formalizing requirements, and organizing work into a prioritized product backlog. The UML artifacts — use case and class diagrams — provided a structural blueprint for the implementation phases, while the architecture and environment sections confirmed the technical choices made. The following chapter translates these specifications into concrete development, beginning with Sprints 1 and 2.

Chapitre

3

Marketplace Foundations

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Introduction

This chapter details the study and implementation of Sprints 1 and 2, which together build the complete public-facing marketplace experience. Sprint 1 establishes the access and identity foundation that governs every subsequent feature, while Sprint 2 opens the platform to users by delivering the asset catalog, the purchasing flow, and the personal download library.

3.1 Sprint 1: Authentication, Profiles and RBAC

Sprint 1 builds the access foundation. It covers user registration, sign-in, password recovery, profile management and the role-based access control model on which every subsequent feature relies.

3.1.1 Sprint 1 Backlog

Table 3.1 summarizes the backlog of the first sprint.

Table 3.1: Sprint 1 Backlog

ID	User Story	Estimation
1	As a visitor, I want to create an account using email/password or OAuth so I can access protected features.	4 days
2	As a registered user, I want to sign in and receive a JWT so I can call protected endpoints.	3 days
3	As a user, I want to reset my password by email so I can recover my account.	2 days
4	As a user, I want to manage my profile so my public page stays up to date.	2 days
5	As an administrator, I want each user to belong to a role that grants a specific set of permissions.	3 days
6	As an administrator, I want sensitive actions to be logged.	2 days

The sprint planning was followed by a conception phase covering the use case model, sequence diagrams and class structure.

3.1.2 Sprint 1 Conception

Use Case Diagram

The figure below illustrates the use case diagram relative to Sprint 1.

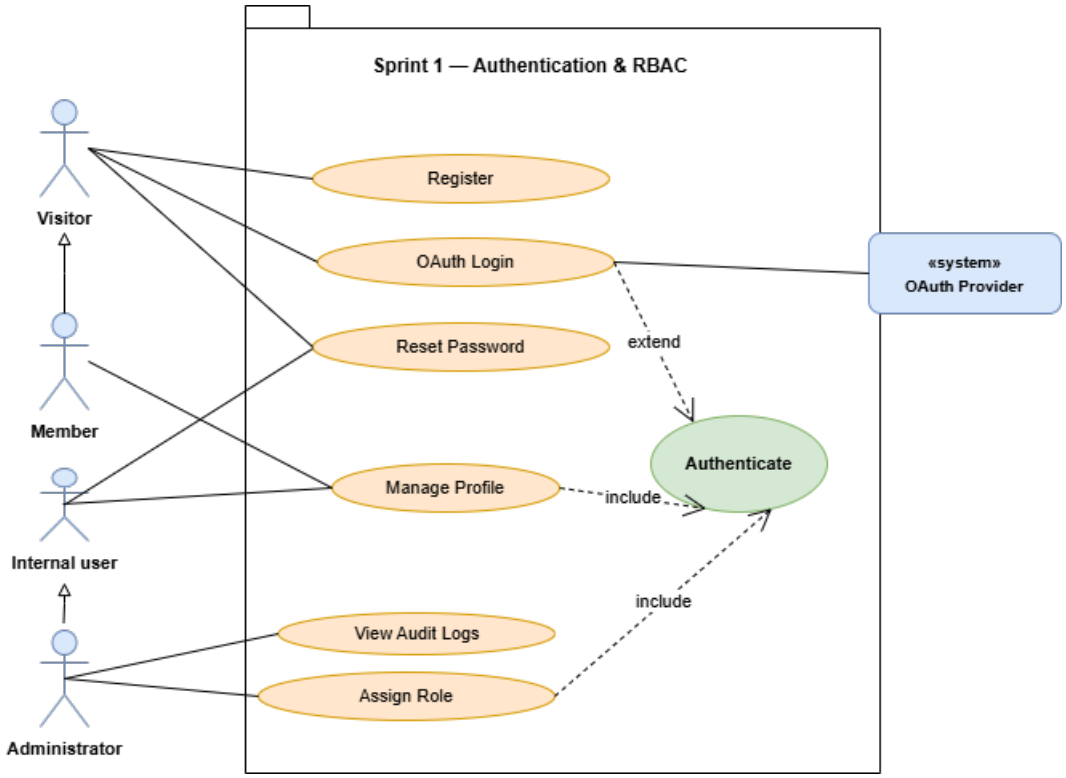

Figure 3.1: Use case diagram — Sprint 1

Table 3.2: Use case description: Authenticate

Use case	Authenticate
Actor	Registered user
Pre-condition	The user has an active account.
Post-condition	The user holds a valid JWT and can access protected routes.
Nominal scenario	1. The user opens the sign-in page. 2. The user enters email and password. 3. The system verifies the credentials. 4. The system signs a JWT and returns it to the browser. 5. The browser stores the token and redirects to the marketplace.
Alternative scenario	3.1. If the credentials are invalid, the system returns an error message. 3.2. Resume at step 2.

Textual description of the use case «Authenticate».

Table 3.3: Use case description: Reset Password

Use case	Reset Password
Actor	Registered user
Pre-condition	The user has lost access to the account but knows the email.
Post-condition	The password is updated and the user can sign in again.
Nominal scenario	1. The user requests a reset link by entering an email. 2. The system generates a reset token and emails the reset link. 3. The user opens the link and submits a new password. 4. The system validates the token and stores the new hashed password. 5. A confirmation message is displayed.
Alternative scenario	4.1. If the token is expired or invalid, an error message is shown. 4.2. The user requests a new reset link.

Textual description of the use case «Reset Password».

Table 3.4: Use case description: Assign Role

Use case	Assign Role
Actor	Administrator
Pre-condition	The administrator is authenticated.
Post-condition	The target user has the chosen role and the associated permissions.
Nominal scenario	1. The administrator opens the user management page. 2. The administrator selects a user. 3. The administrator chooses a role from the list. 4. The system updates the role assignment and logs the action. 5. A success message is displayed.
Alternative scenario	3.1. If the administrator does not have the permission to assign the chosen role, an error is shown. 3.2. The administrator picks a different role.

Textual description of the use case «Assign Role».

Sequence Diagrams

Sequence diagram «Authenticate». When a user attempts to authenticate, the browser sends the credentials to the `AuthController`. The system looks up the user record, verifies the password hash and signs a JWT. The token is returned to the browser, which stores it and attaches it as a Bearer header on every subsequent call. If credentials are invalid, the system returns an error and no token is issued.

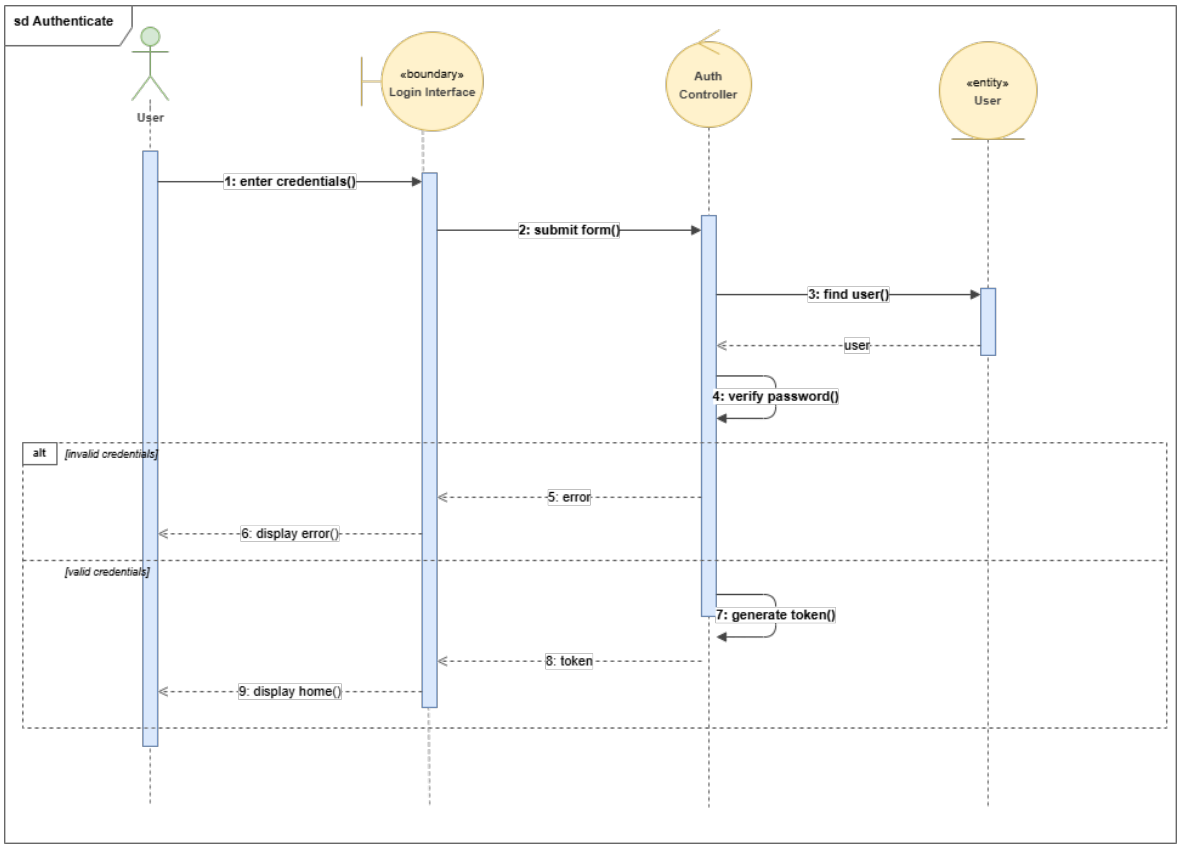

Figure 3.2: Sequence diagram «Authenticate»

Sequence diagram «Reset Password». When a user requests a password reset, the system generates a unique token, stores it on the user record with an expiry, and sends a reset link by email. When the user opens the link and submits a new password, the system verifies the token, hashes the new password and clears the reset token. If the token is missing or expired, the system rejects the request.

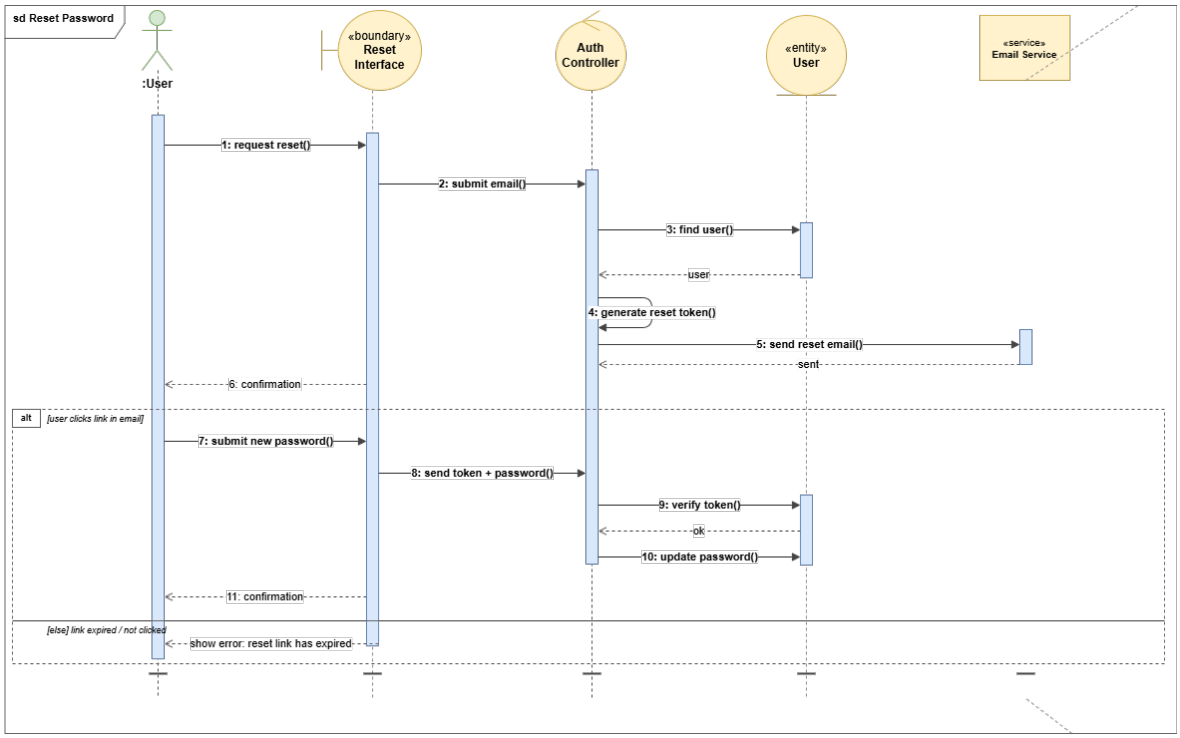

Figure 3.3: Sequence diagram «Reset Password»

Class Diagram

The class diagram of Sprint 1 centres on the `User` entity linked to the RBAC trio `Role`, `Permission` and `AuditLog`. A user holds one or more roles, each role carries a set of permissions, and every sensitive action is traced in the audit log.

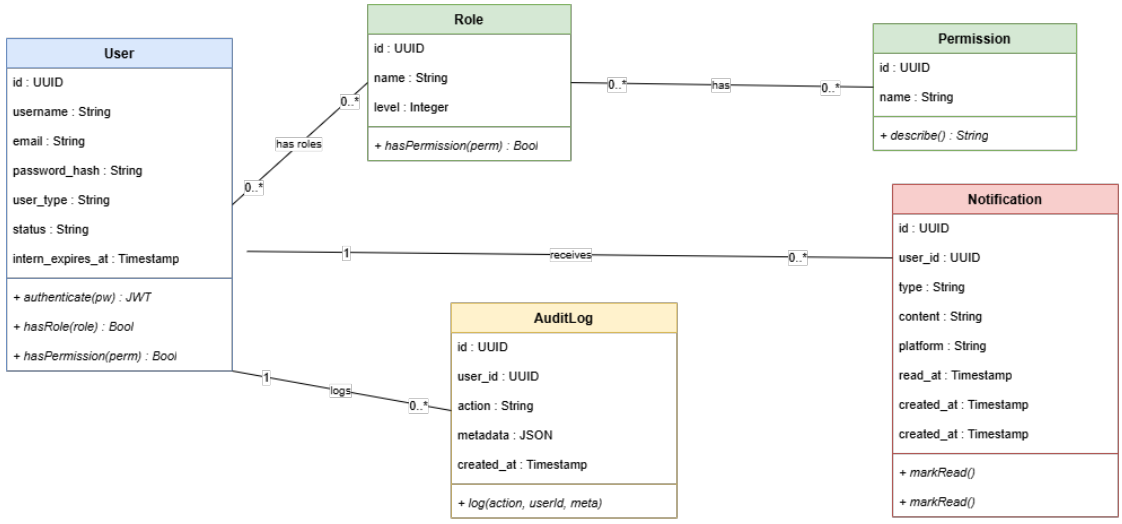

Figure 3.4: Class diagram — Sprint 1

The conceptual model described above guided the implementation, whose key interfaces are presented in the following subsection.

3.1.3 Sprint 1 Realization

In this section we present the interfaces developed during Sprint 1.

«Sign-in interface». This interface allows the user to authenticate securely through email and password or through an OAuth provider. On success, the system stores the JWT and redirects to the marketplace.

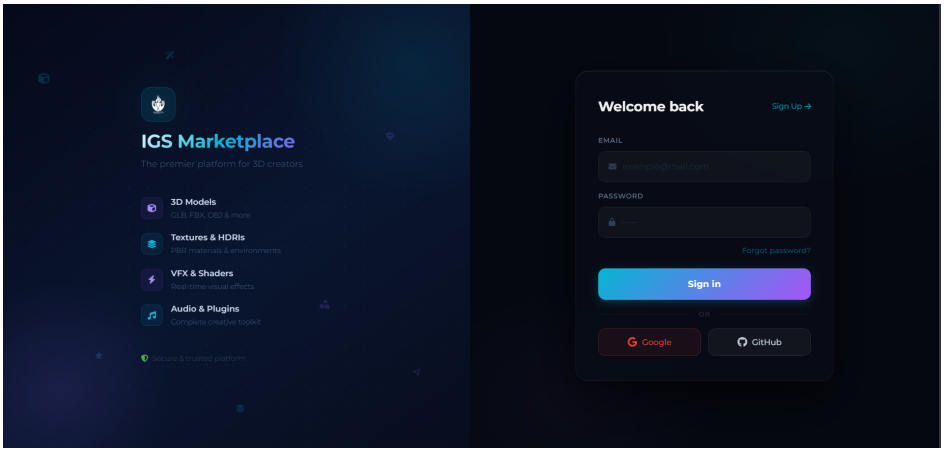

Figure 3.5: Interface «Sign in»

«Sign-up interface». This interface lets a new visitor create an account by filling in a short form. Validation is enforced on the email format and the password strength.

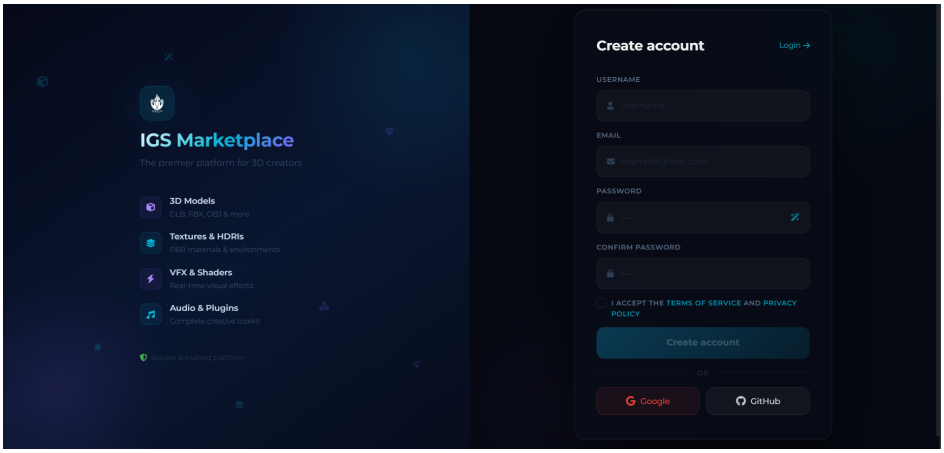

Figure 3.6: Interface «Sign up»

«Profile management». This interface allows the authenticated user to view and update personal information (avatar, display name, bio, contact details) and to change the account password.

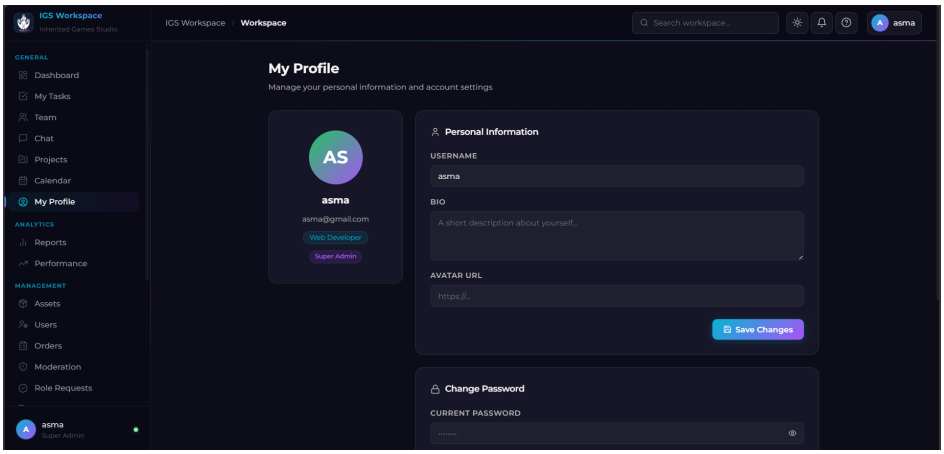

Figure 3.7: Interface «Profile»

The following subsection summarizes the outcomes achieved at the end of Sprint 1.

3.1.4 Sprint 1 Conclusion

At the end of Sprint 1 the platform has a working authentication layer with JWT and OAuth, a profile management page, and a hierarchical RBAC model with six roles. This foundation conditions every feature delivered in the following sprints.

Building on the secure foundation established in Sprint 1, Sprint 2 focuses on delivering the public-facing marketplace experience.

3.2 Sprint 2: Catalog, Cart, Payment and Library

Sprint 2 turns the platform into a usable storefront. It delivers the filterable asset catalog, the interactive 3D and WebXR previews, the shopping cart, the Stripe and Flouci payment flows and the personal download library.

3.2.1 Sprint 2 Backlog

Table 3.5 summarizes the backlog of the second sprint.

Table 3.5: Sprint 2 Backlog

ID	User Story	Estimation
1	As a visitor, I want to browse a paginated catalogue with filters by category, price and tags.	3 days
2	As a visitor, I want to consult an asset detail page with a 3D preview adapted to the asset type.	4 days
3	As a member, I want to enter VR / immersive mode for compatible 3D assets so I can inspect them at scale.	2 days
4	As a member, I want to add or remove assets from a personal cart.	2 days
5	As a member, I want to pay for the cart through Stripe or Flouci and receive instant access.	4 days
6	As a member, I want a personal library listing all the assets I can download, with signed-URL downloads.	3 days

The conception phase for Sprint 2 establishes the use case model, sequence diagrams and class structure for the catalog and purchase features.

3.2.2 Sprint 2 Conception

Use Case Diagram

The figure below illustrates the use case diagram relative to Sprint 2.

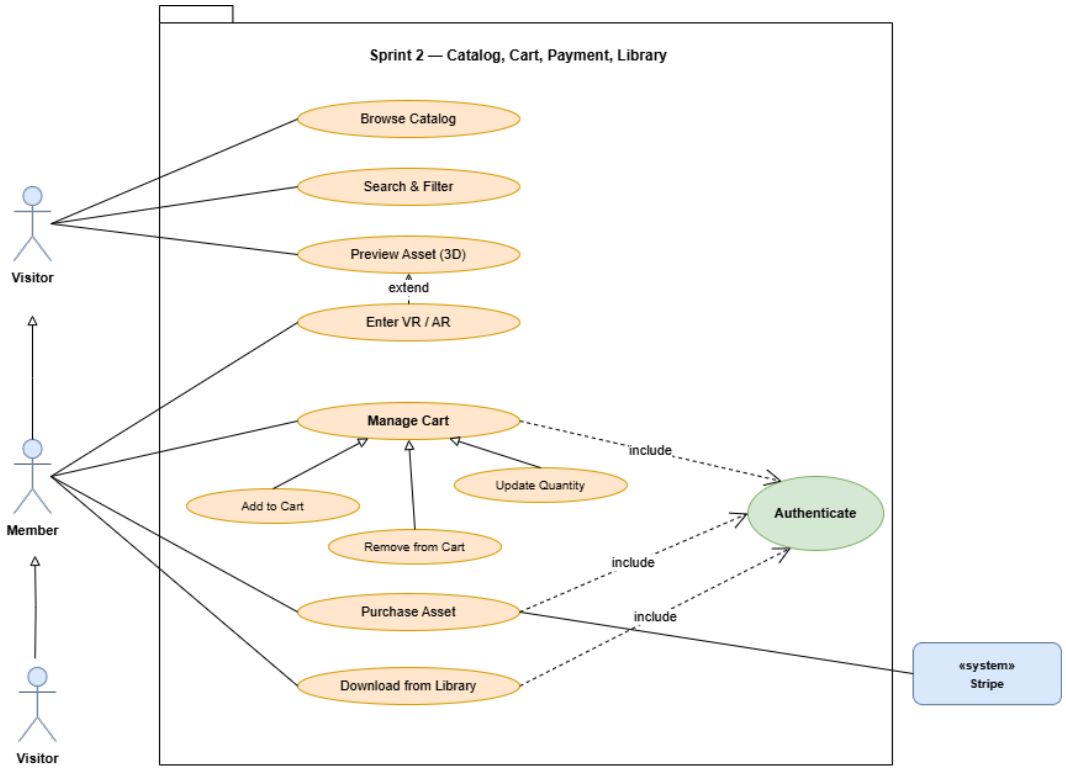

Figure 3.8: Use case diagram — Sprint 2

Table 3.6: Use case description: Browse and Filter Catalog

Use case	Browse and Filter Catalog
Actor	Visitor / Member
Pre-condition	The catalog page is accessible.
Post-condition	The catalog displays the filtered list of approved assets.
Nominal scenario	1. The visitor opens a category page. 2. The visitor applies filters on price, rating or tags. 3. The system queries the catalog with the selected filters. 4. The system returns a paginated list of matching assets. 5. The visitor clicks an asset card to open the detail page.
Alternative scenario	4.1. If no asset matches the filters, an empty-state message is displayed.

Textual description of the use case «Browse and Filter Catalog».

Table 3.7: Use case description: Purchase an Asset

Use case	Purchase an Asset
Actor	Member
Pre-condition	The member is authenticated and has at least one asset in the cart.
Post-condition	The order is recorded as paid and the asset is available in the library.
Nominal scenario	1. The member opens the cart and clicks Checkout. 2. The member selects Stripe or Flouci as payment method. 3. The system creates a payment intent and presents the card form. 4. The payment gateway confirms the transaction. 5. The system records the order, copies the cart items and unlocks the library. 6. A confirmation message is shown.
Alternative scenario	4.1. If the payment fails, the order stays pending and the library is not unlocked. 4.2. The member can retry from the cart.

Textual description of the use case «Purchase an Asset».

Table 3.8: Use case description: Download from Library

Use case	Download from Library
Actor	Member
Pre-condition	The member has at least one asset unlocked in the library.
Post-condition	The asset file is downloaded to the user's machine.
Nominal scenario	1. The member opens the library page. 2. The member clicks Download on an asset. 3. The system re-checks the access level for the asset. 4. The system generates a signed URL with a one-hour validity. 5. The browser fetches the file directly from cloud storage.
Alternative scenario	3.1. If the access level is blocked, the system returns a 403 error.

Textual description of the use case «Download from Library».

Sequence Diagrams

Sequence diagram «Purchase an Asset». When a member proceeds to checkout, the backend creates a payment intent at the gateway and returns the client secret to the browser. The user confirms the card and the gateway notifies the backend of success. The backend then opens a transaction, inserts the order, copies the cart items into `order_items` with the locked commission split, grants library access, and returns the confirmation to the browser.

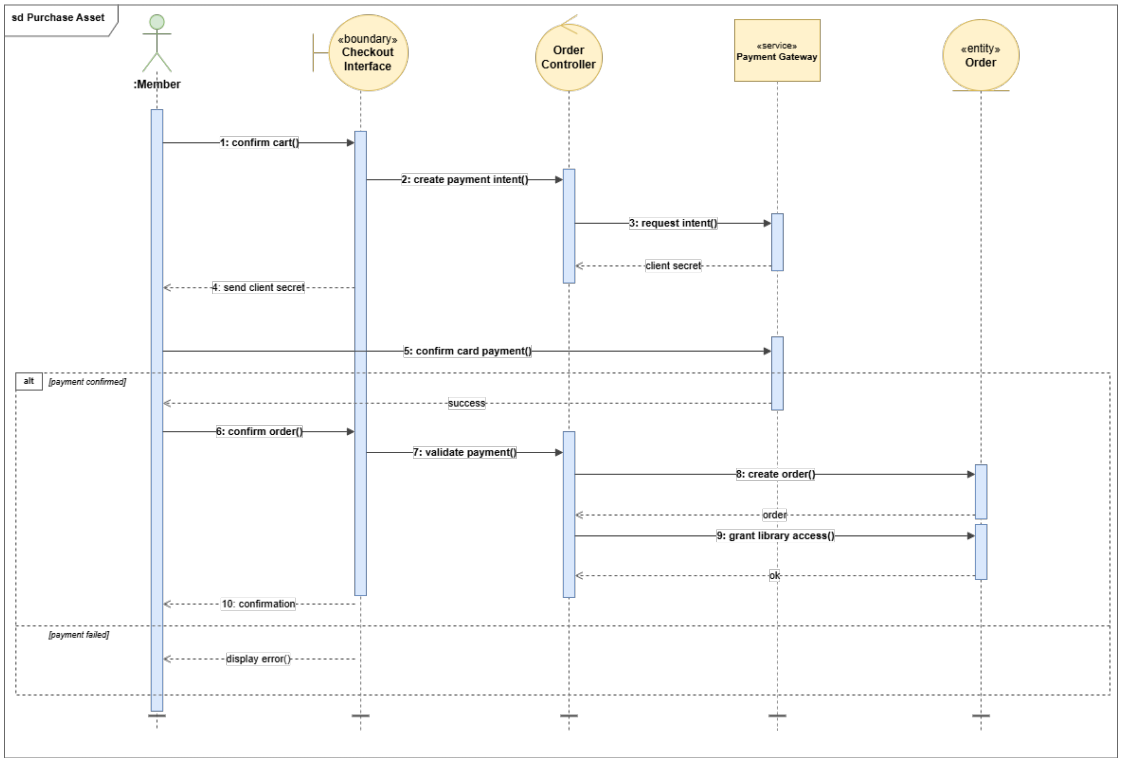

Figure 3.9: Sequence diagram «Purchase an Asset»

Sequence diagram «Download from Library». When a member clicks Download, the backend re-checks the access level for the asset. If the access is granted, the backend retrieves the file path and the bucket and generates a one-hour signed URL through the storage service. The browser uses this signed URL to fetch the file directly from the CDN, without exposing the bucket path.

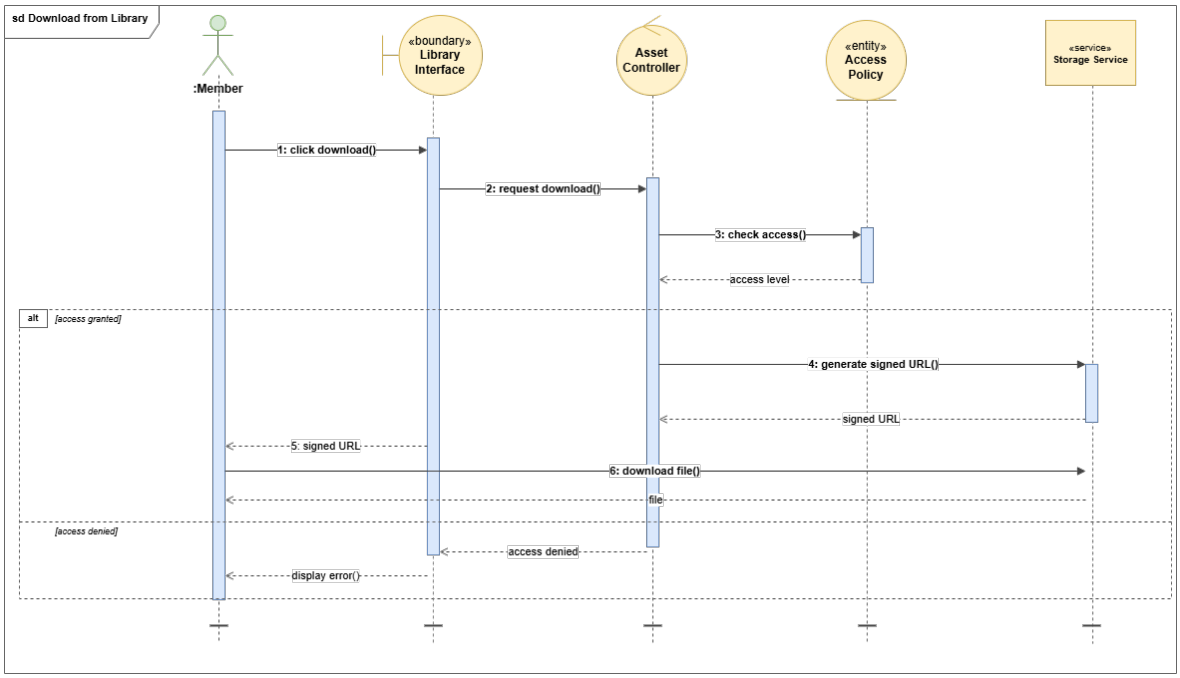

Figure 3.10: Sequence diagram «Download from Library»

Class Diagram

The class diagram of Sprint 2 introduces the `Asset` and `Category` entities alongside the purchase pipeline: `CartItem`, `Order`, `OrderItem` and `UserDownload`. Each order item locks the price and commission split at purchase time so revenue figures remain stable.

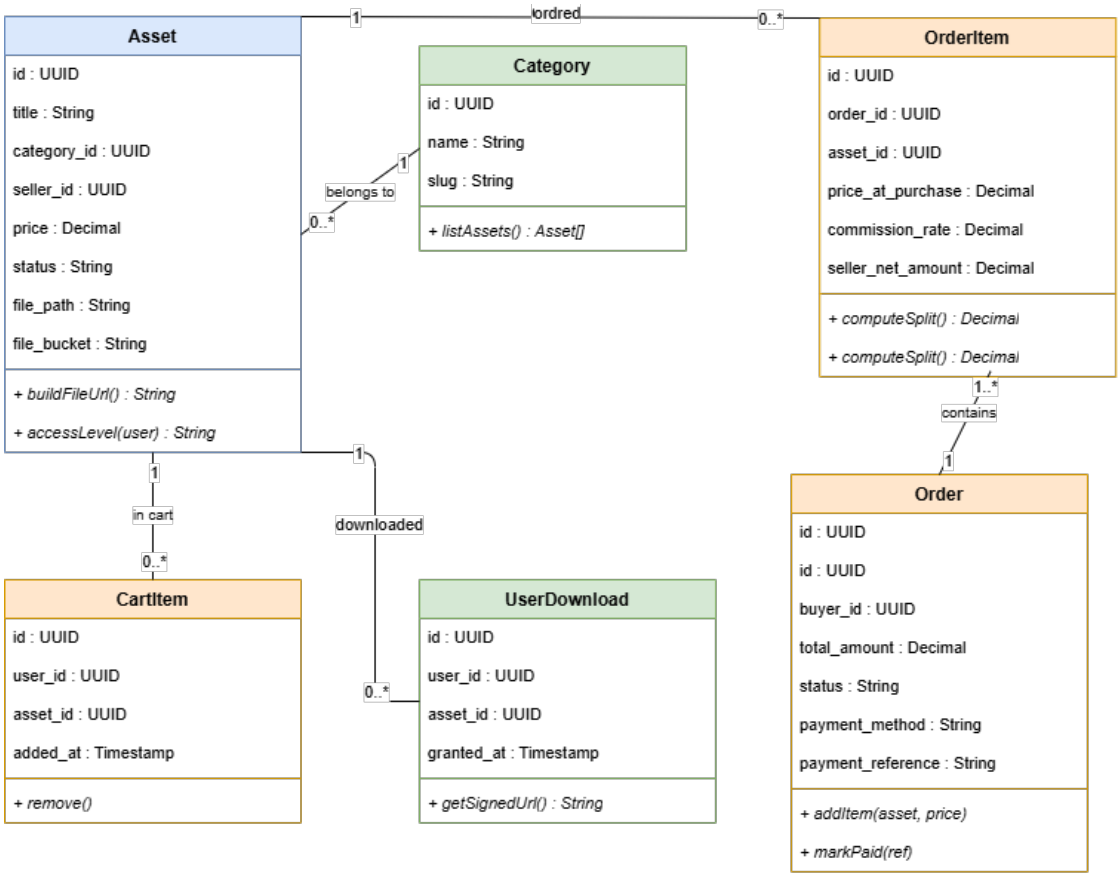

Figure 3.11: Class diagram — Sprint 2

The following interfaces implement the catalog, cart and library features designed in the conception phase.

3.2.3 Sprint 2 Realization

In this section we present the interfaces developed during Sprint 2.

«Catalog page». This interface lets visitors browse one of the thirteen asset categories. A sidebar exposes filters by price, rating, tags and file format. Each card shows the thumbnail, the seller, the price and an interactive preview adapted to the asset type.

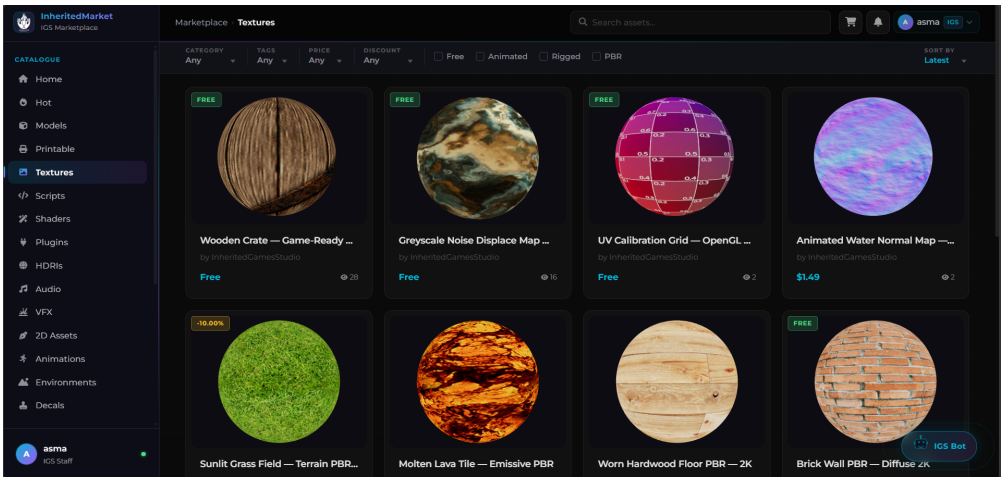

Figure 3.12: Interface «Catalog»

«Catalog filters». The filter sidebar lets visitors narrow results by price range, rating, tags and file format. Selections update the catalog grid instantly without a page reload.

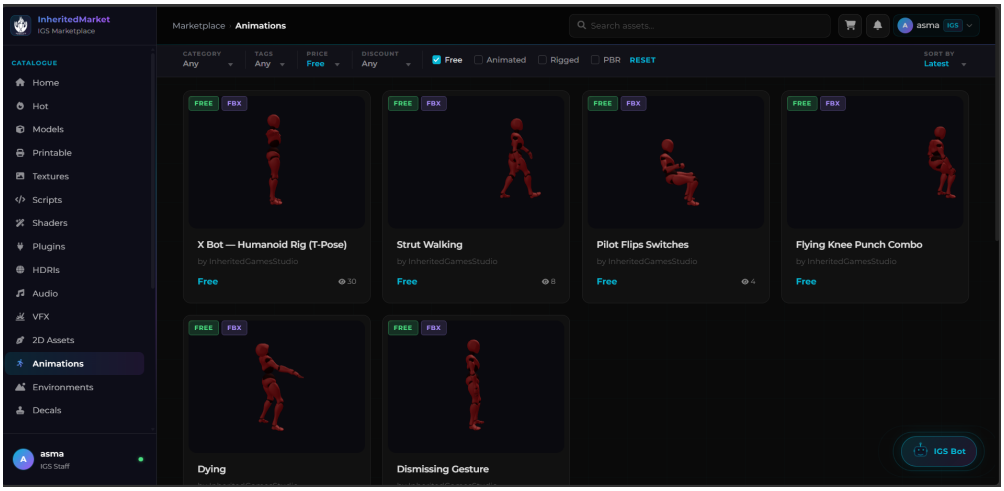

Figure 3.13: Interface «Catalog filters»

«Shopping cart». This interface lists the assets selected by the member. The user can remove an item or update the quantity, then proceed to checkout with Stripe or Flouci.

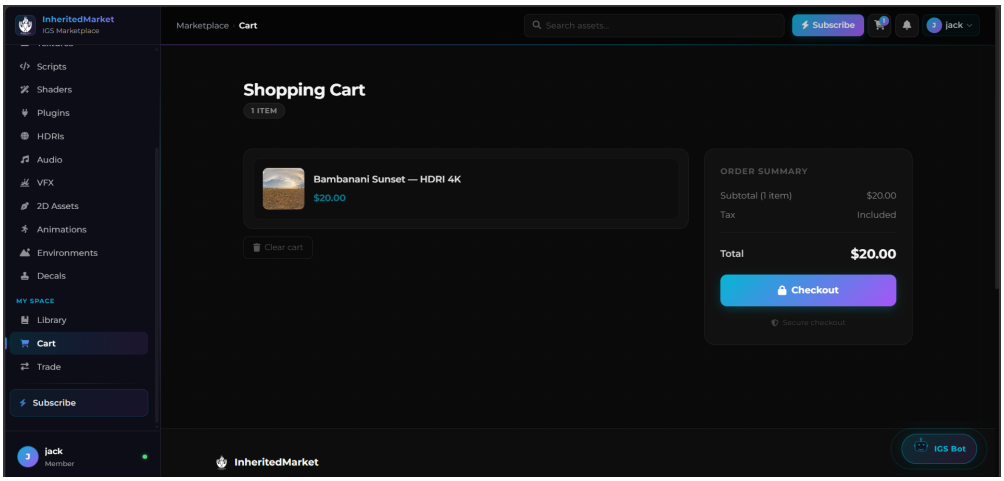

Figure 3.14: Interface «Cart»

«Library». This interface gathers all the assets the member has unlocked (purchased, gifted, voucher-redeemed or granted by IGS). Each row exposes a Download button that requests a one-hour signed URL from the backend.

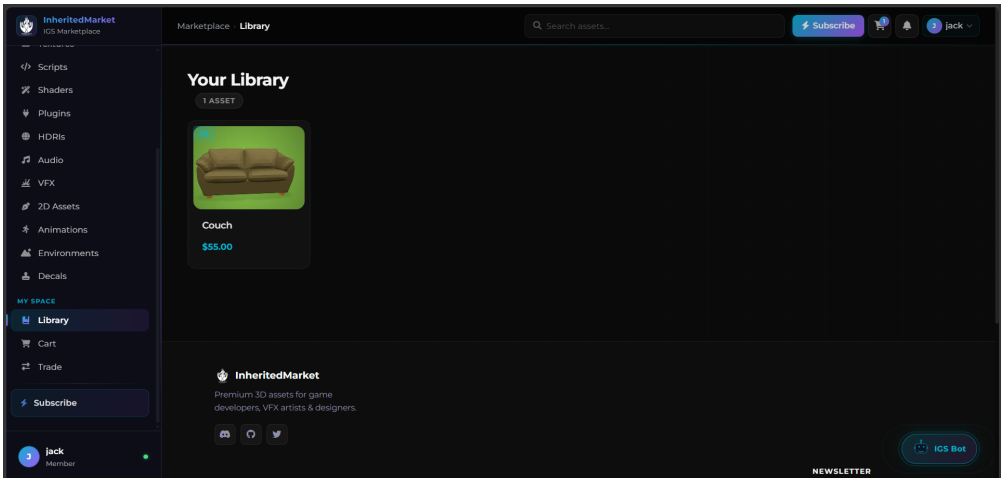

Figure 3.15: Interface «Library»

The following subsection summarizes the achievements at the end of Sprint 2.

3.2.4 Sprint 2 Conclusion

At the end of Sprint 2 the public marketplace is fully functional. Visitors can browse and preview assets in 3D, members can purchase through Stripe or Flouci, and the personal library exposes time-limited signed downloads.

Conclusion

This chapter translated the Sprint 1 and Sprint 2 backlog items into a working marketplace foundation. The authentication and RBAC layer established the security model relied upon by every subsequent feature, while Sprint 2 delivered the full public browsing, purchasing, and download experience. The following chapter builds on this foundation by addressing the seller workflow, content moderation, and the internal workspace.

Chapitre

4

Seller Workflow and Workspace

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Introduction

This chapter covers the study and implementation of Sprints 3 and 4, which extend the platform beyond the public marketplace into seller-facing and internal tooling. Sprint 3 closes the content production loop by enabling sellers, administrators, and buyers to interact around the asset catalog. Sprint 4 then delivers the IGS Workspace, an internal dashboard designed for team collaboration, project management, and platform administration.

4.1 Sprint 3: Asset Upload, Moderation and Reviews

Sprint 3 closes the production loop of the marketplace. It covers the seller on-boarding workflow, the asset upload and moderation pipeline, the review and rating system, the peer-to-peer trade offers and the Creator Studio dashboard.

4.1.1 Sprint 3 Backlog

Table 4.1 summarizes the backlog of the third sprint.

Table 4.1: Sprint 3 Backlog

ID	User Story	Estimation
1	As a member, I want to request seller access with a justification so I can publish assets.	2 days
2	As a seller, I want to upload assets and submit them for moderation.	4 days
3	As an admin, I want a moderation queue where I can approve or reject submitted assets.	3 days
4	As a seller, I want to request edits on already-approved assets without bypassing moderation.	2 days
5	As a buyer, I want to leave a rating and a written review on owned assets.	2 days
6	As a user, I want to be notified about request, order and moderation events.	2 days
7	As a seller, I want a creator studio with stats and an asset table.	3 days
8	As a user, I want to propose peer-to-peer trades of assets I own against assets I want.	2 days

The conception phase models the seller workflow, moderation pipeline and review system for this sprint.

4.1.2 Sprint 3 Conception

Use Case Diagram

The figure below illustrates the use case diagram relative to Sprint 3.

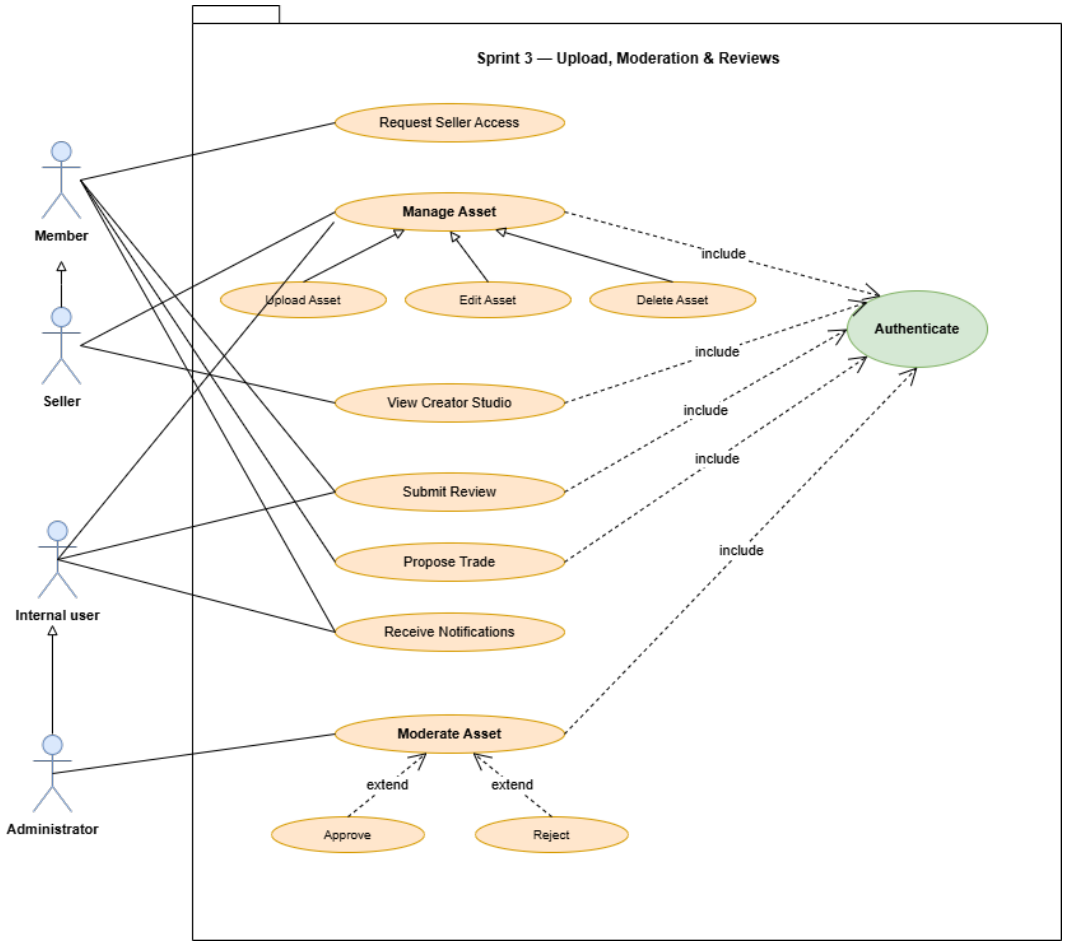

Figure 4.1: Use case diagram — Sprint 3

Table 4.2: Use case description: Upload an Asset

Use case	Upload an Asset
Actor	Seller
Pre-condition	The seller is authenticated and holds the seller role.
Post-condition	The asset is stored as pending and enters the moderation queue.
Nominal scenario	1. The seller opens the upload form. 2. The seller selects a thumbnail and an asset file. 3. The seller fills in the metadata (title, category, tags, price). 4. The system validates the file and stores it in cloud storage. 5. The asset record is created with status pending. 6. A success message is displayed.
Alternative scenario	4.1. If the file size or the MIME type is invalid, an error message is shown. 4.2. The seller selects a valid file and resumes at step 2.

Textual description of the use case «Upload an Asset».

Table 4.3: Use case description: Moderate an Asset

Use case	Moderate an Asset
Actor	Administrator
Pre-condition	A submitted asset is waiting in the moderation queue.
Post-condition	The asset is approved and published, or rejected with a note.
Nominal scenario	1. The administrator opens the moderation queue. 2. The administrator expands a pending asset. 3. The administrator inspects the file, the thumbnail and the metadata. 4. The administrator chooses Approve or Reject and adds an optional note. 5. The system updates the status, logs the action and notifies the seller.
Alternative scenario	4.1. If the administrator rejects, the seller is notified with the note and can resubmit.

Textual description of the use case «Moderate an Asset».

Table 4.4: Use case description: Submit a Review

Use case	Submit a Review
Actor	Member
Pre-condition	The member owns the asset.
Post-condition	The review and the rating are saved and visible on the asset page.
Nominal scenario	1. The member opens the detail page of an owned asset. 2. The member selects a star rating and types a written review. 3. The system verifies the ownership and stores the review. 4. The system runs sentiment analysis on the text. 5. The new review is displayed on the asset page.
Alternative scenario	3.1. If the member does not own the asset, the review form is disabled.

Textual description of the use case «Submit a Review».

Sequence Diagrams

Sequence diagram «Upload an Asset». When a seller submits the upload form, the browser sends the thumbnail and the asset file to the upload service. The system validates the MIME type at the byte level and stores each file in the correct cloud bucket. Once both files are saved, the controller creates the asset record with status pending. The seller is then notified that the submission is awaiting moderation.

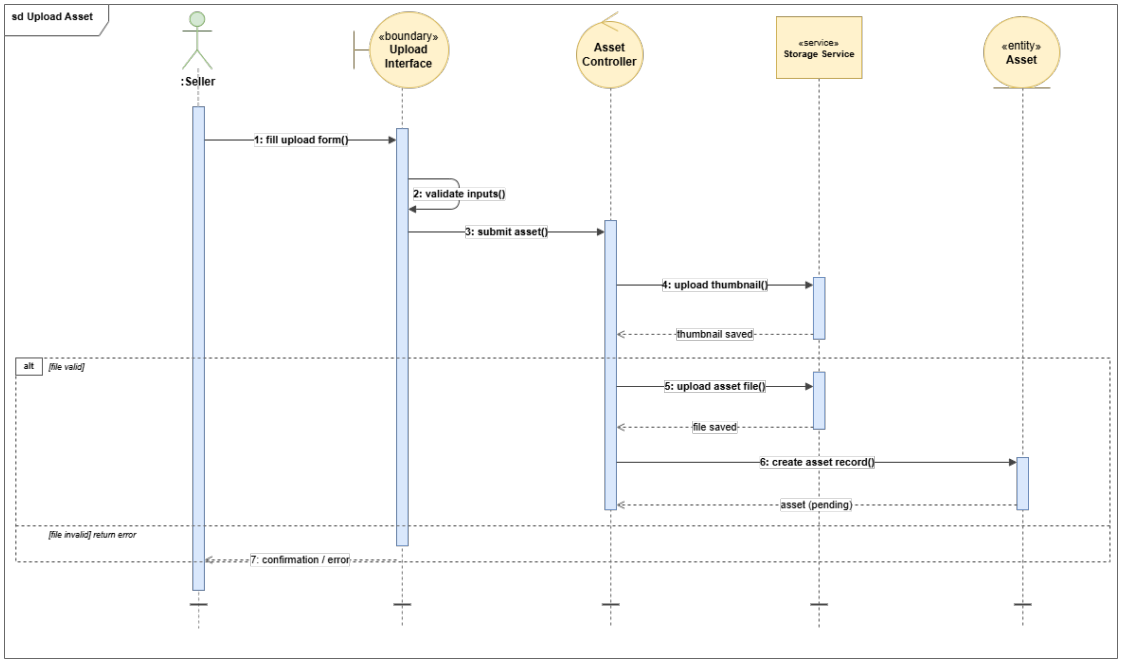

Figure 4.2: Sequence diagram «Upload an Asset»

Sequence diagram «Moderate an Asset». When an administrator opens the moderation queue, the system loads all submissions with status pending. After reviewing the file and the metadata, the administrator issues a decision. The system updates the asset status, writes an entry to the audit log and notifies the seller through the notification service.

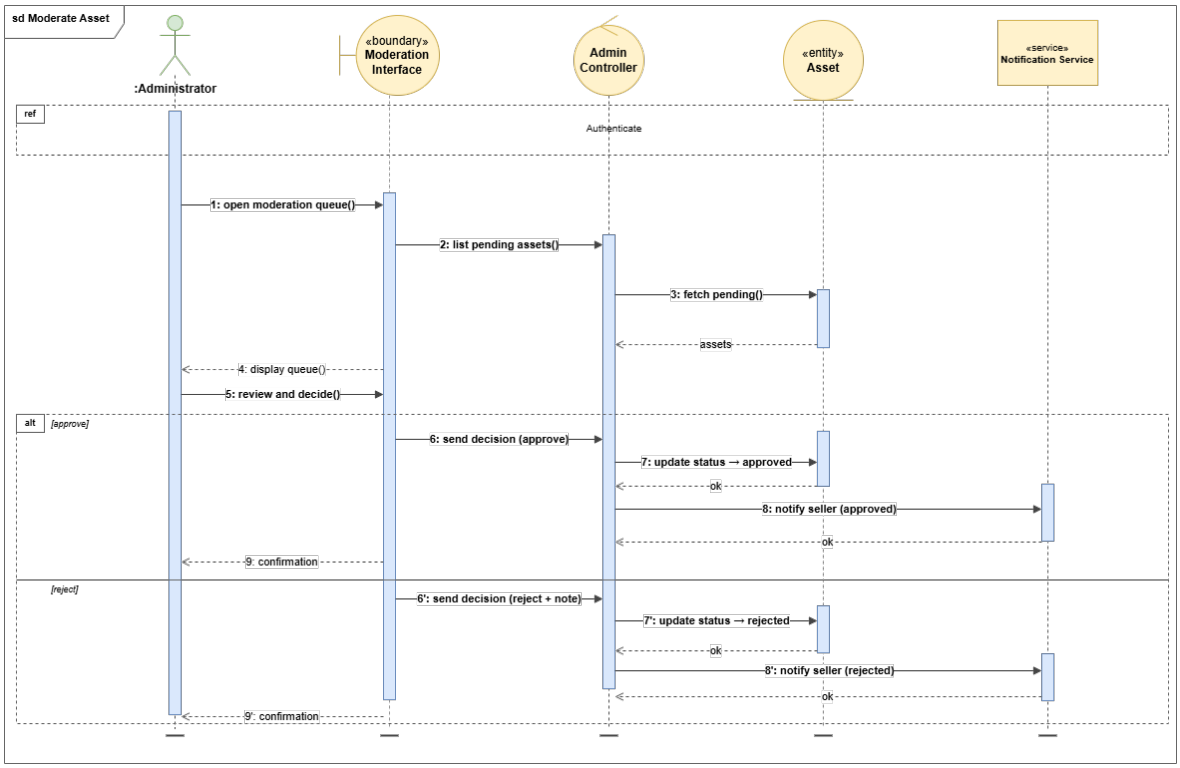

Figure 4.3: Sequence diagram «Moderate an Asset»

Class Diagram

The class diagram of Sprint 3 extends the `Asset` entity with a status state machine and connects it to `AssetEditRequest`, `Review` (with an `ai_sentiment` column filled in Sprint 5), `Trade` and `Subscription`.

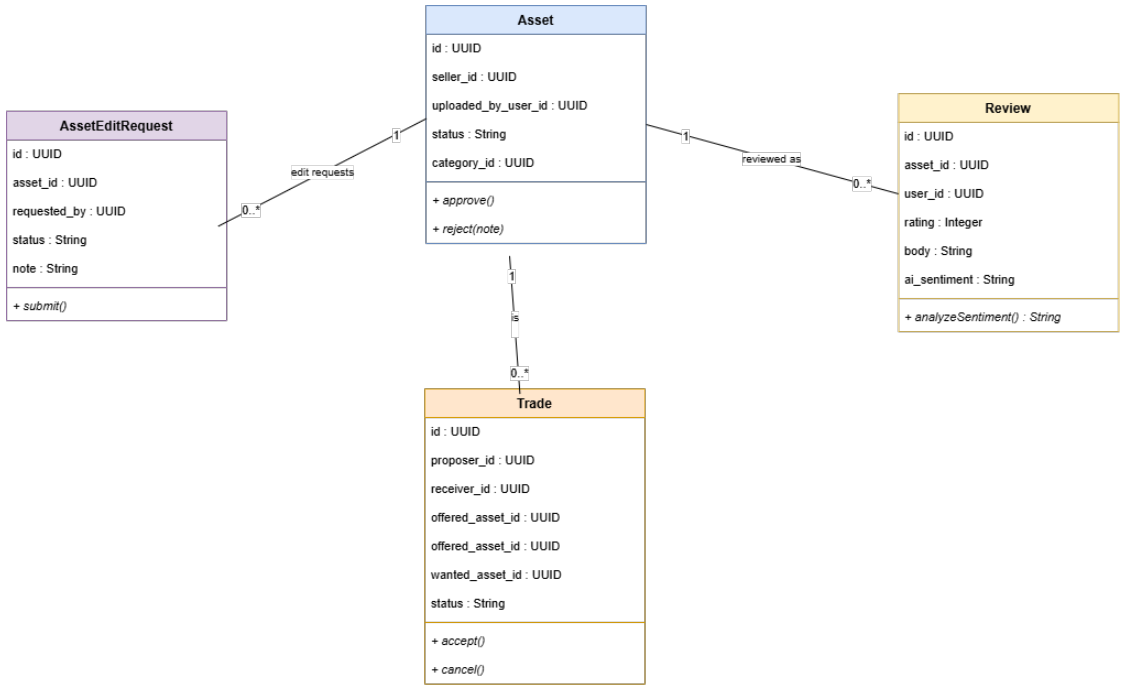

Figure 4.4: Class diagram — Sprint 3

The following interfaces implement the seller, moderation and review features defined in the conception phase.

4.1.3 Sprint 3 Realization

In this section we present the interfaces developed during Sprint 3.

«Seller profile». This interface gathers the public information of a seller: avatar, biography, published assets, total sales and aggregated rating. Visitors can open it from any asset card.

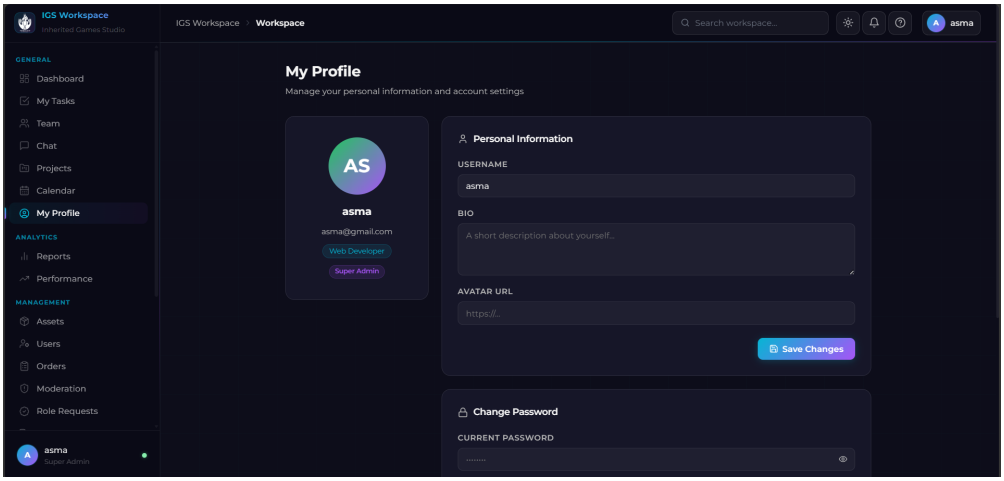

Figure 4.5: Interface «Seller profile»

«Asset reviews». This interface shows the chronological list of written reviews and ratings for an asset. Each card displays the rating, the author, the body, and a sentiment badge produced by the AI service introduced in Sprint 5.

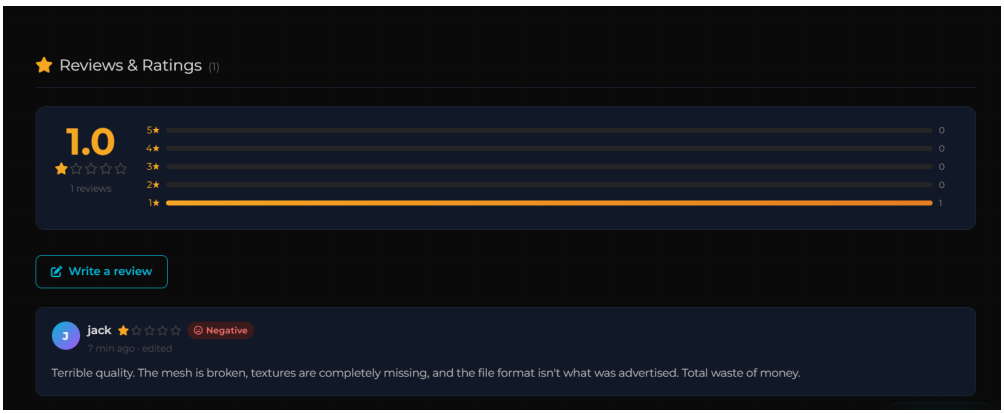

Figure 4.6: Interface «Reviews»

«Creator Studio». This interface gives sellers a personal performance dashboard: total revenue, active listings, recent sales and a paginated asset table with per-row

statistics.

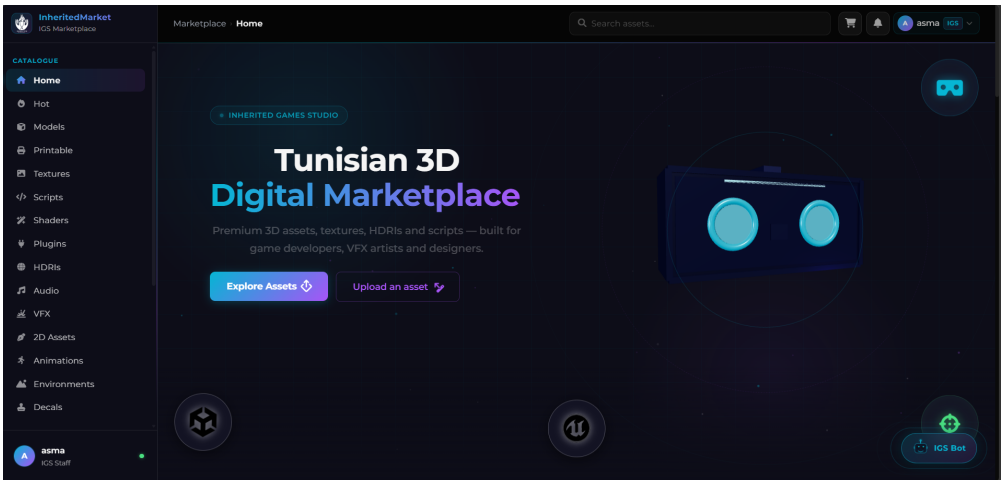

Figure 4.7: Interface «Creator Studio»

«Moderation queue». This interface lets administrators consult every pending submission. Each row is expandable to show the asset metadata, the file preview, the seller note and the decision buttons.

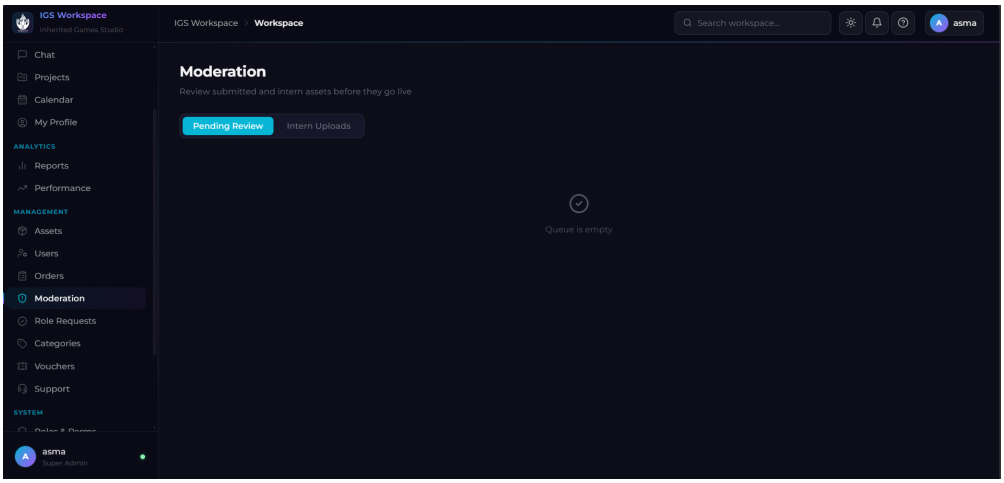

Figure 4.8: Interface «Moderation queue»

The following subsection consolidates the achievements of Sprint 3.

4.1.4 Sprint 3 Conclusion

Sprint 3 closes the production loop. Sellers can publish assets, administrators control the catalog through moderation, buyers can rate and review, and notifications keep every actor informed of state changes.

With the marketplace production loop complete, Sprint 4 shifts focus to the internal workspace and administrative tooling.

4.2 Sprint 4: IGS Workspace and Administration

Sprint 4 delivers the internal IGS Workspace. It covers project and task management, the team chat, the calendar, the moderation of intern uploads and intern access requests, the administrative panel for users and categories, and the system pages reserved for super-administrators.

4.2.1 Sprint 4 Backlog

Table 4.5 summarizes the backlog of the fourth sprint.

Table 4.5: Sprint 4 Backlog

ID	User Story	Estimation
1	As an internal user, I want a workspace with role-gated navigation.	3 days
2	As an intern, I want a Kanban task board with priorities and assignees.	3 days
3	As an internal user, I want to create and manage projects with members and a date range.	2 days
4	As an internal user, I want a calendar that shows task deadlines and project milestones.	1 day
5	As an internal user, I want a real-time chat.	3 days
6	As an employee, I want analytics reports on revenue, sales and per-team performance.	2 days
7	As an admin, I want to manage users, role requests, categories and vouchers.	3 days
8	As an admin, I want to approve or reject intern asset access requests.	2 days
9	As a super-admin, I want to consult audit logs and configure the platform.	2 days

The conception phase for Sprint 4 models the workspace collaboration, administration and audit features.

4.2.2 Sprint 4 Conception

Use Case Diagram

The figure below illustrates the use case diagram relative to Sprint 4.

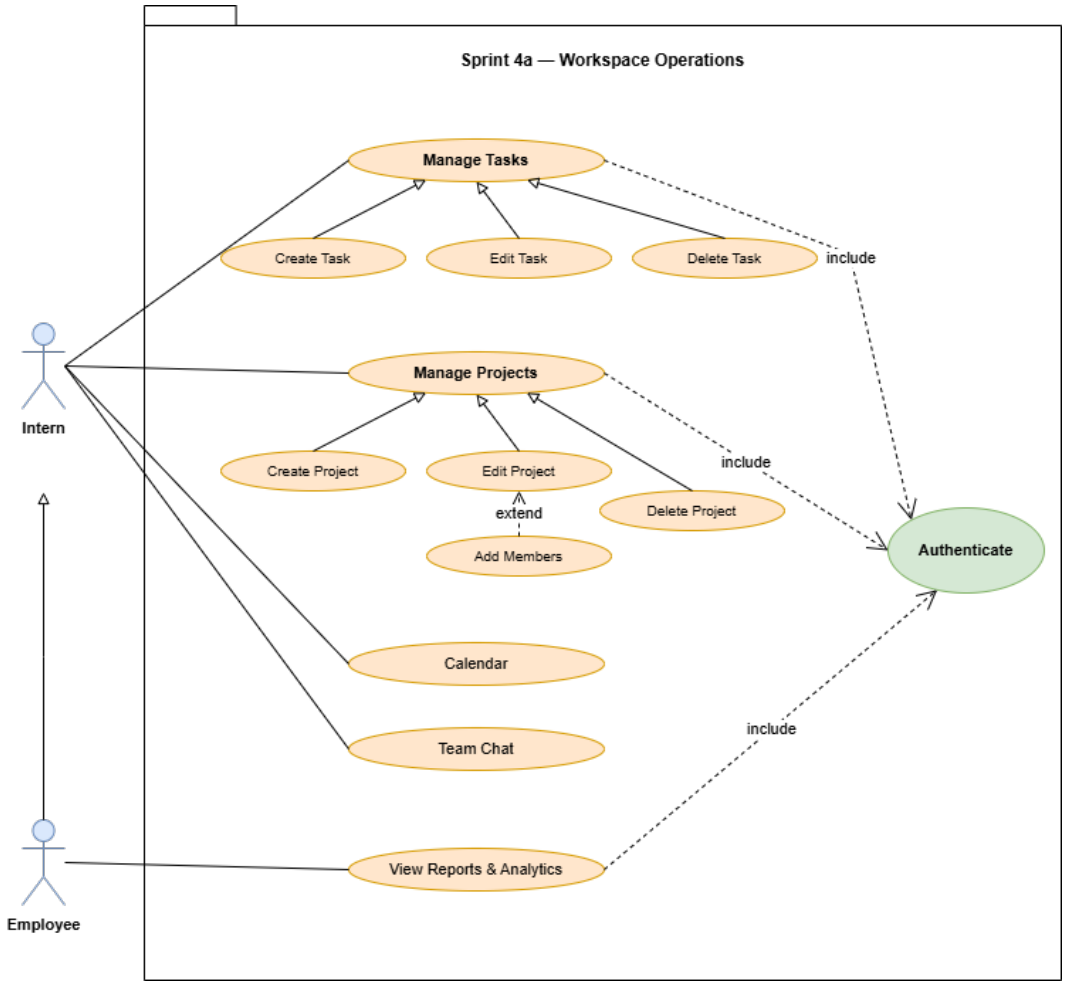

Figure 4.9: Use case diagram — Sprint 4 (Part A)

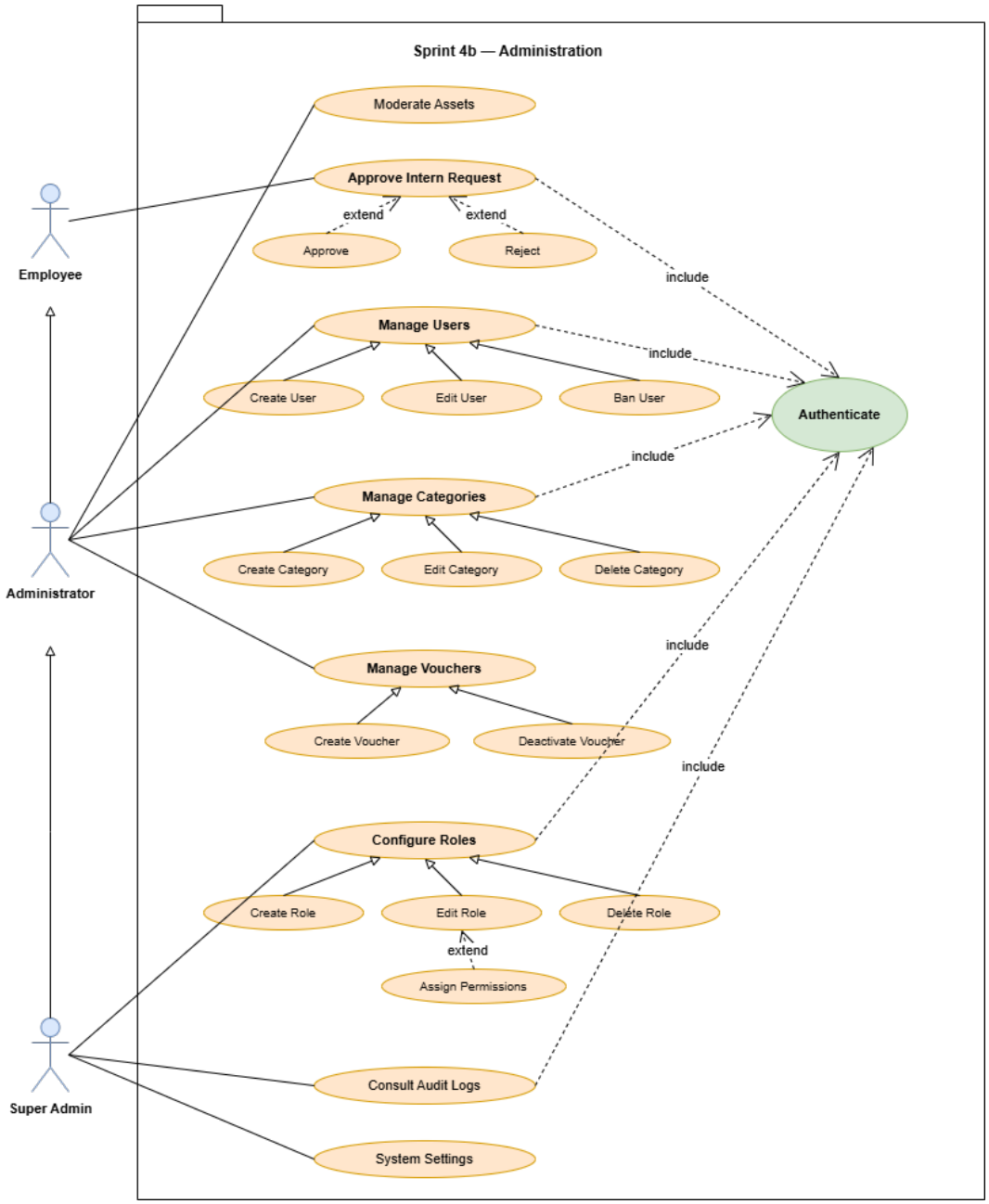

Figure 4.10: Use case diagram — Sprint 4 (Part B)

Table 4.6: Use case description: Create a Task

Use case	Create a Task
Actor	Internal user (Intern, Employee, Admin)
Pre-condition	The user is authenticated in the workspace.
Post-condition	The task is created on the Kanban board and the assignees are notified.
Nominal scenario	1. The user opens the project board. 2. The user clicks New Task. 3. The user fills in the title, the description, the priority and the assignees. 4. The system creates the task with status To-do. 5. The system notifies the assignees through Socket.IO. 6. The task appears on the Kanban board.
Alternative scenario	3.1. If the user is not a member of the project, an error message is shown.

Textual description of the use case «Create a Task».

Table 4.7: Use case description: Approve Intern Access Request

Use case	Approve Intern Access Request
Actor	Employee / Administrator
Pre-condition	An intern submitted a request for a paid IGS asset.
Post-condition	The request is approved and the asset is added to the intern's library, or rejected with a note.
Nominal scenario	1. The employee opens the workspace moderation page. 2. The employee selects a pending intern request. 3. The employee chooses Approve or Reject and adds a note. 4. The system updates the request status. 5. On approval, the asset is added to the intern's library. 6. The intern is notified of the decision.
Alternative scenario	3.1. If the employee rejects, the intern is notified and may submit a new request.

Textual description of the use case «Approve Intern Access Request».

Table 4.8: Use case description: Consult Audit Logs

Use case	Consult Audit Logs
Actor	Super-administrator
Pre-condition	The super-administrator is authenticated in the workspace.
Post-condition	The filtered list of audit entries is displayed.
Nominal scenario	1. The super-administrator opens the audit log page. 2. The super-administrator applies filters on action type or user. 3. The system queries the audit log table. 4. The system returns a paginated list of entries. 5. The super-administrator expands an entry to inspect the JSON metadata.
Alternative scenario	4.1. If no entry matches the filter, an empty-state message is displayed.

Textual description of the use case «Consult Audit Logs».

Sequence Diagrams

WebSockets and Real-time Communication. The workspace team chat and live task notifications are built on top of **WebSockets**, a full-duplex communication protocol that keeps a persistent TCP connection open between the client and the server. Unlike the polling model, where the browser repeatedly queries the server for new data, WebSockets allow the server to *push* events to connected clients instantly without waiting for a request.

The platform uses **Socket.IO** as the WebSocket layer. Socket.IO extends the raw WebSocket protocol with automatic reconnection, room-based broadcasting (one message delivered to all members of a named room) and a fallback to HTTP long-polling for environments that block WebSocket connections. When an internal user connects to the workspace, the server authenticates their JWT and registers them in a personal room keyed to their user ID; users are also joined into project rooms and team-chat channel rooms. Any event emitted to a room (a new message, a task status change, an incoming notification) is delivered in real time to all currently connected members of that room.

Sequence diagram «Create a Task». When an internal user submits a new task, the browser sends the form to the workspace controller. The controller verifies the project membership, inserts the task in the database with status To-do and writes a

notification row for each assignee. The system then emits a Socket.IO event so the assignees see the task appear in real time on their Kanban board.

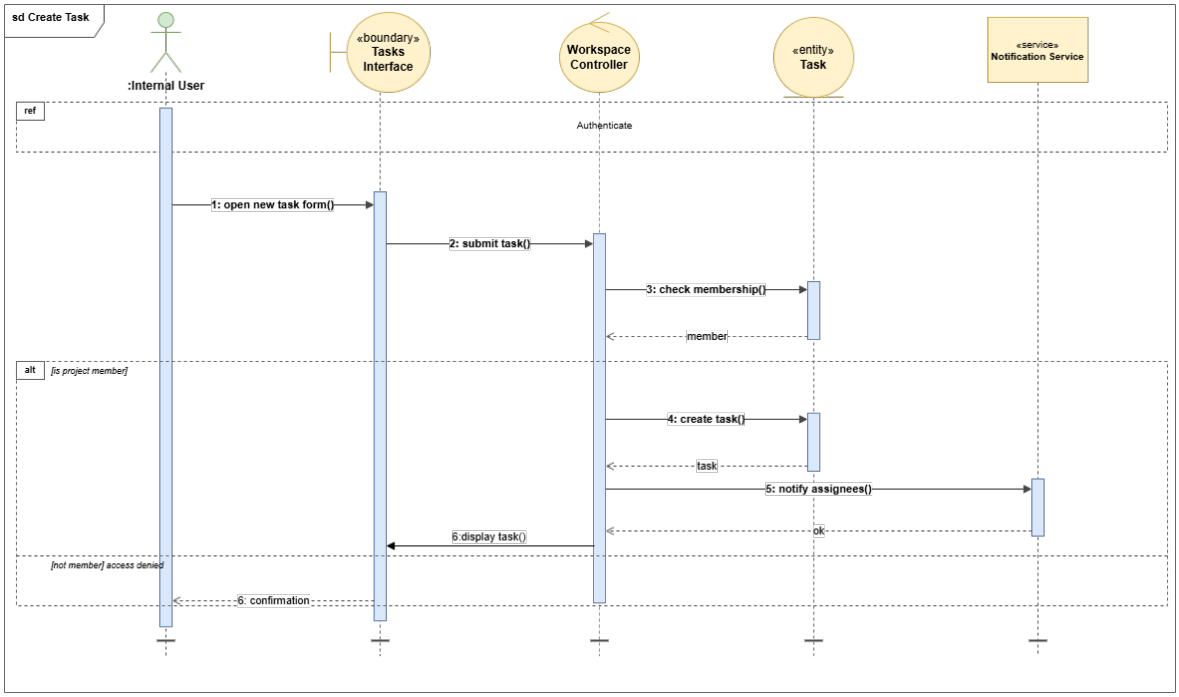

Figure 4.11: Sequence diagram «Create a Task»

Sequence diagram «Approve Intern Access Request». When an intern requests access to a paid IGS asset, the system stores the request and notifies the supervisors. When the supervisor opens the workspace moderation page and approves the request, the system adds the asset to the intern's library and sends a notification with the decision back to the intern.

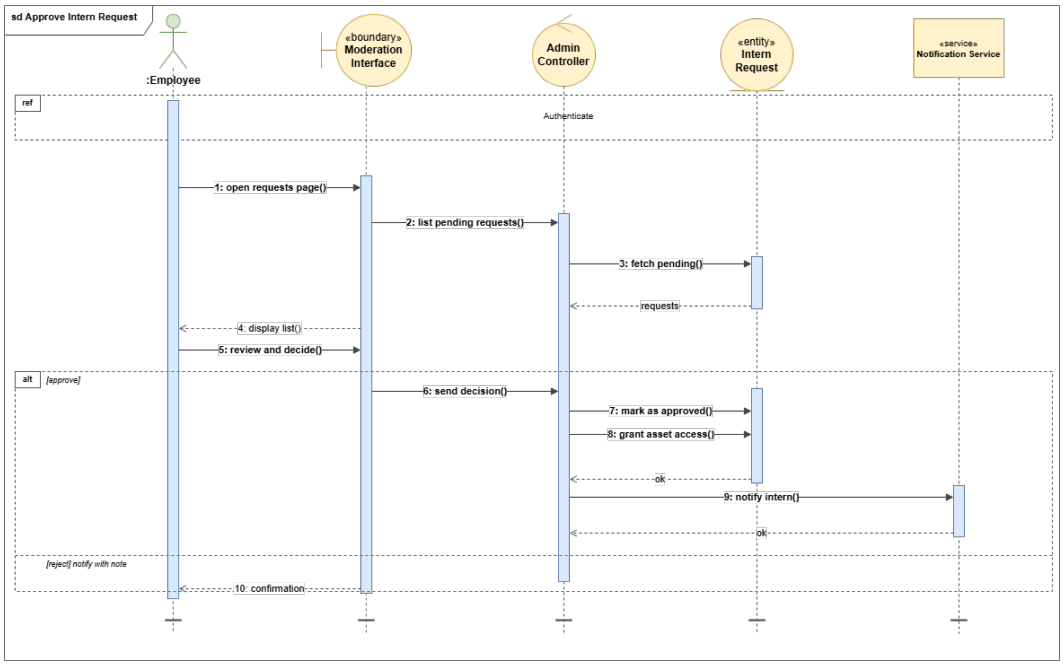

Figure 4.12: Sequence diagram «Approve Intern Access Request»

Class Diagram

The class diagram of Sprint 4 covers the internal collaboration layer: `Project` and `WorkspaceTask` for project tracking, `ChatChannel` and `Message` for real-time communication, `InternAssetRequest` for supervised access and `Voucher` for the discount code management.

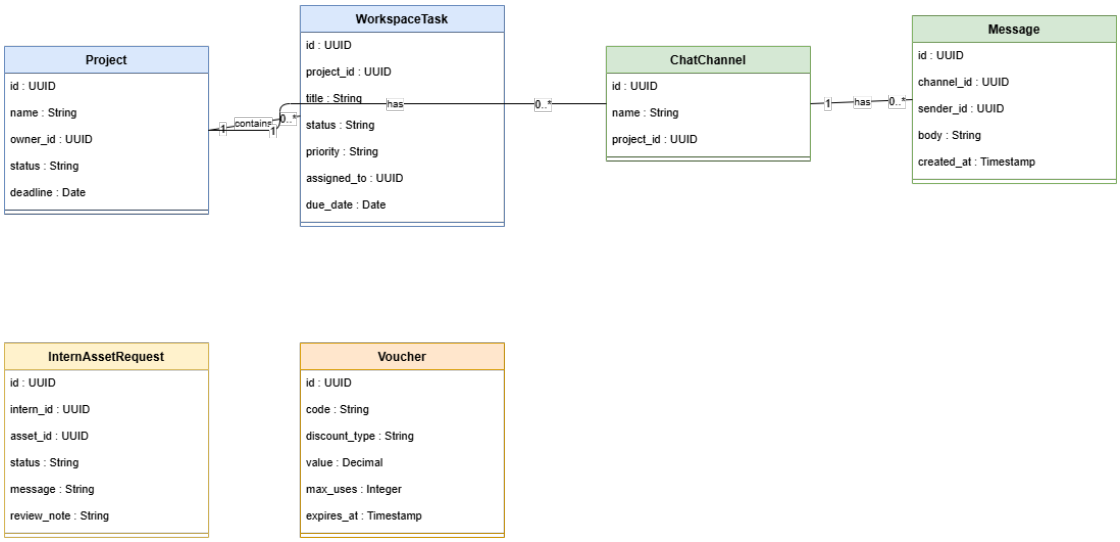

Figure 4.13: Class diagram — Sprint 4

The interfaces developed during Sprint 4 are presented in the following subsection.

4.2.3 Sprint 4 Realization

In this section we present the interfaces developed during Sprint 4.

«Workspace dashboard». This interface is the landing page of the workspace. It centralises the team's recent activity: open tasks, ongoing projects, pinned notifications and quick access to the chat.

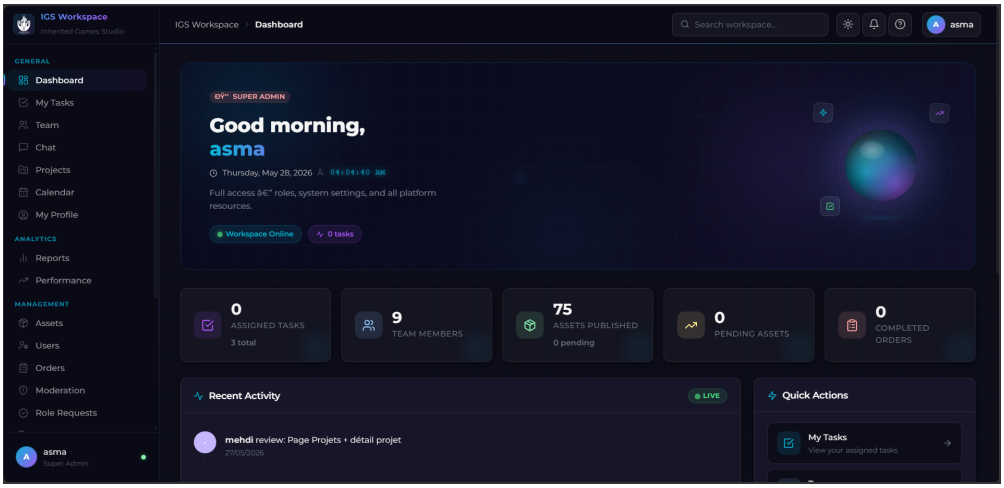

Figure 4.14: Interface «Workspace dashboard»

«Kanban task board». This interface displays tasks organised in four columns (To-do, In progress, Review, Done). Each card carries a priority badge, an assignee avatar and a due date.

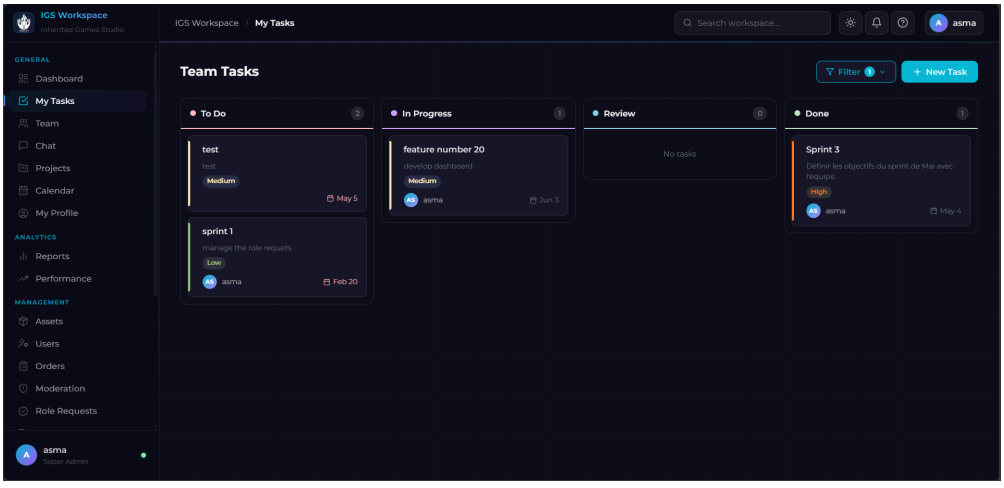

Figure 4.15: Interface «Tasks»

«Team chat». This interface provides real-time messaging backed by Socket.IO. Internal users can exchange messages across shared channels or in direct conversations.

New messages appear instantly without page reload, and a persistent scroll history is loaded from the database on first connection.

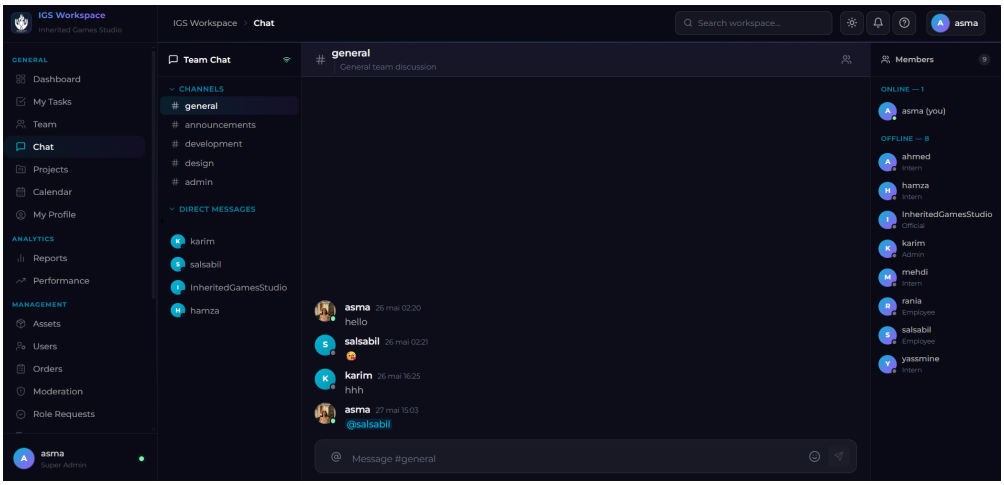

Figure 4.16: Interface «Team chat»

«User management». This interface lists every platform user. The administrator can change the role, edit the profile or ban an account.

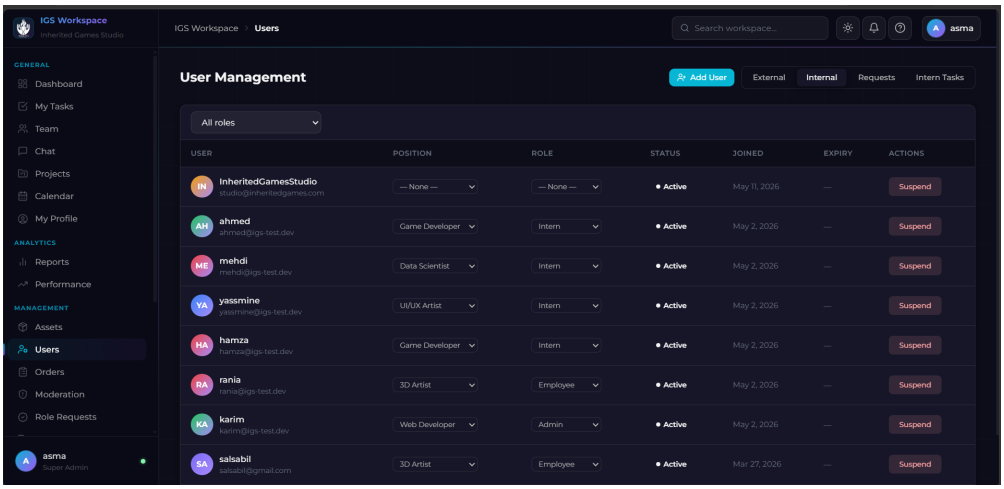

Figure 4.17: Interface «User management»

«Order management». This interface displays every order placed on the platform with the payment status, the buyer, the amount and the commission. The administrator can refund an order or inspect the line items.

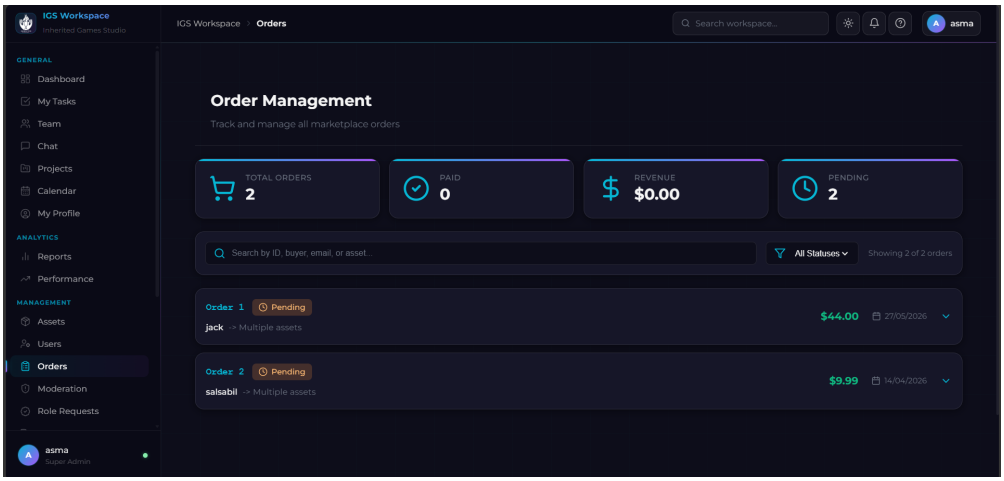

Figure 4.18: Interface «Order management»

«Roles and permissions». This interface lists every role with its current permission set and the live user count. Super-administrators can edit the role-permission assignment.

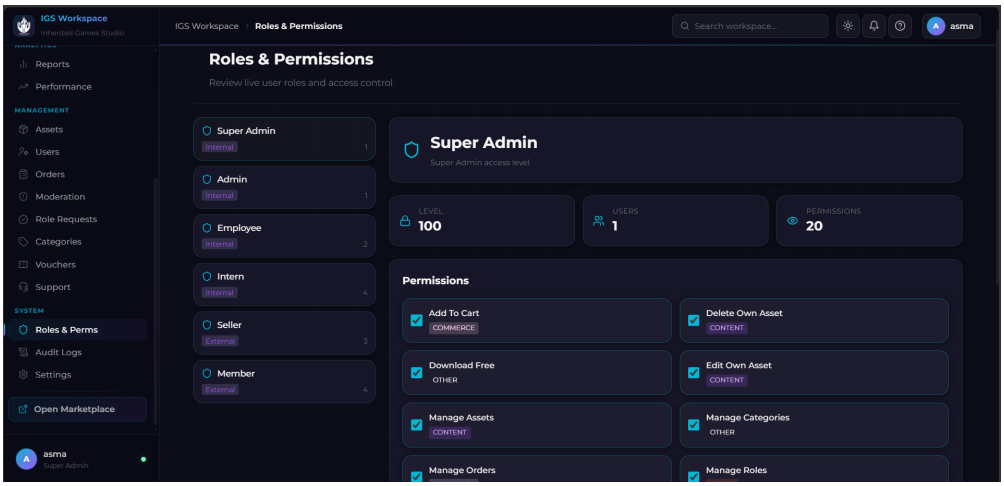

Figure 4.19: Interface «Roles and permissions»

«Audit logs». This interface lets the super-administrator filter the audit log by action and by user, browse paginated results and expand any entry to inspect its JSON metadata.

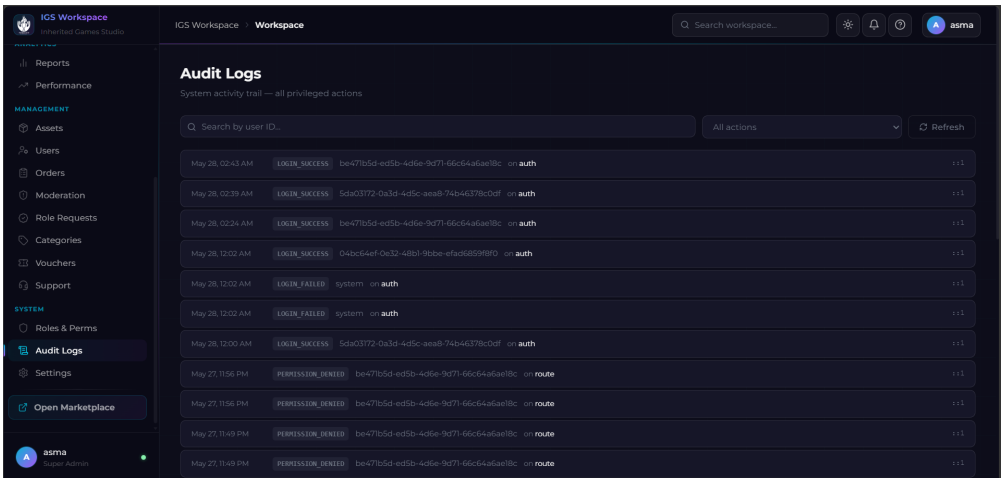

Figure 4.20: Interface «Audit logs»

«System settings». This interface gathers the platform-wide configuration. The most critical entry is the commission rate applied at order time.

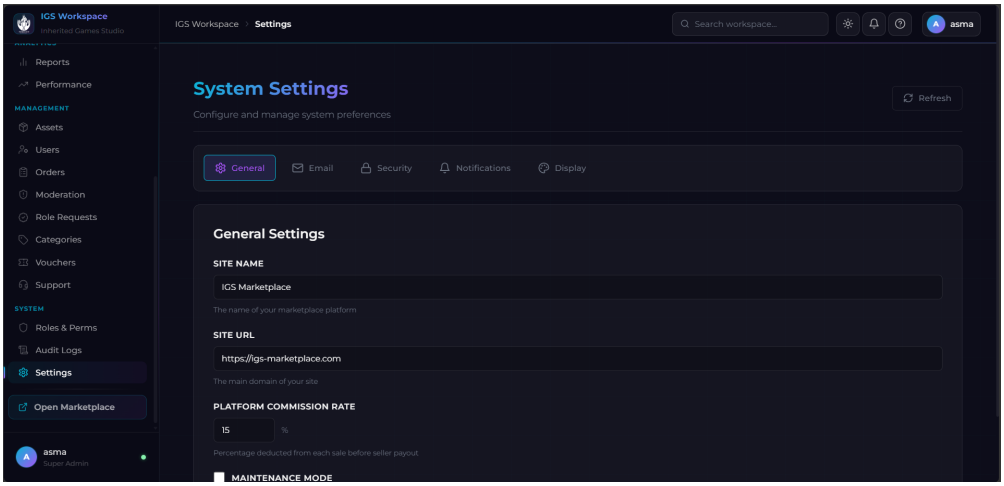

Figure 4.21: Interface «System settings»

«Support inbox». This interface centralises the support tickets opened by sellers. The administrator can filter by status, send a message and close the ticket.

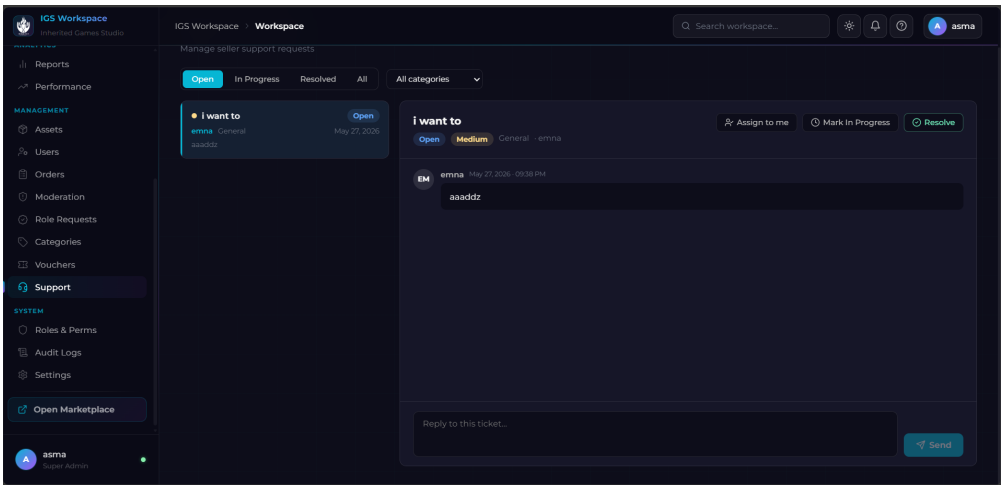

Figure 4.22: Interface «Support inbox»

The following subsection summarizes the outcomes of Sprint 4.

4.2.4 Sprint 4 Conclusion

At the end of Sprint 4 the studio team has a fully functional internal ERP. Daily operations are centralised, moderators have a single queue for external and intern submissions, and super-administrators control the platform from a dedicated system menu while every sensitive action is traced through the audit log.

Conclusion

This chapter demonstrated how Sprints 3 and 4 transformed the platform from a public marketplace into a complete, end-to-end managed ecosystem. The seller workflow, moderation pipeline, and review system established the content governance model, while the internal Workspace gave the IGS team the operational tools needed for daily management and oversight. The following chapter completes the platform by presenting the AI-powered services, immersive VR/AR preview, and production deployment infrastructure.

Chapitre

5

Advanced Features

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Introduction

This chapter presents the advanced capabilities that distinguish the IGS Marketplace from a conventional asset store: AI-powered services integrated into the user experience, security hardening mechanisms protecting the platform against common vulnerabilities, and an immersive WebXR preview feature complemented by a production deployment infrastructure.

5.1 AI Services

Three AI services enrich the platform with intelligent behaviour. Each was selected to match the available infrastructure and the nature of the data: a large language model for conversational assistance, a pre-trained transformer for sentiment classification, and a deterministic similarity algorithm for personalized recommendations.

The first AI service is a conversational assistant integrated into every page of the marketplace.

5.1.1 AI Chatbot

Analysis

Overview. The chatbot is a conversational assistant integrated into every page of the marketplace. It allows users to search the catalog, ask platform-related questions and retrieve order information in natural language. The system operates through a three-stage pipeline: intent classification, optional catalog data retrieval, and response generation. This architecture keeps latency low by routing simple requests locally and only invoking the language model when a free-form answer is required.

Stage 1 — Intent Classification with TF-IDF and Linear SVM. Intent classification determines the purpose of a user message before any further processing. The platform uses a two-step approach: statistical vectorization followed by a linear classifier.

TF-IDF — Term Frequency–Inverse Document Frequency

TF-IDF [21] is a statistical measure that quantifies the importance of each word in a message relative to the entire training corpus. Words that appear frequently in the message but rarely across all messages receive high scores, making them strong discriminators. Formally, for a word w in document d from corpus D:

$$\text{TF-IDF}(w, d, D) = \frac{f_{w,d}}{\sum_{w' \in d} f_{w',d}} \times \log\left(\frac{|D|}{|\{d' \in D : w \in d'\}|}\right)$$

Each message is transformed into a sparse numerical vector representing its lexical content.

Support Vector Machine

A **Support Vector Machine** [22] then classifies the TF-IDF vector. SVMs find the optimal separating hyperplane between training classes by maximizing the margin between the closest examples of each class (the *support vectors*). The linear variant is particularly effective for text classification tasks, since TF-IDF feature spaces are generally linearly separable and the linear kernel trains efficiently on high-dimensional sparse data [23].

The classifier recognizes six intent categories, illustrated in Table 5.1.

Table 5.1: Intent categories handled by the classifier

Intent	Example utterance
Product search	"Show me animated character models under $10"
Platform help	"How do I upload my first asset?"
Order status	"Where is my download?"
Seller information	"How do commissions work?"
Greeting	"Hello, what can you help me with?"
Out of scope	"What is the weather today?"

Stage 2 — Retrieval-Augmented Generation. When a product-search intent is detected, the platform queries the asset catalog using keywords from the message and injects the matching results into the model's context window. This **retrieval-augmented generation** (RAG) pattern [24] grounds the model's response in real catalog data and eliminates the risk of the model fabricating non-existent products.

Stage 3 — Response Generation with a Large Language Model. The final response is generated by **Qwen2.5-72B-Instruct** [25], a large language model developed by the Qwen Team at Alibaba Cloud.

Transformer Architecture and Large Language Models

Large language models are built on the **Transformer** architecture [26], which relies on a *self-attention mechanism* allowing every token in the input sequence to attend to every other token simultaneously, capturing long-range dependencies without recurrence. Pre-trained on massive text corpora and subsequently fine-tuned with instruction-following data, these models generate fluent, contextually coherent answers conditioned on a provided system prompt and conversation history.

The model is served through the Hugging Face Inference API [20], eliminating the need for local GPU infrastructure. A rule-based fallback is activated whenever the external service is unreachable, ensuring the assistant remains functional at all times.

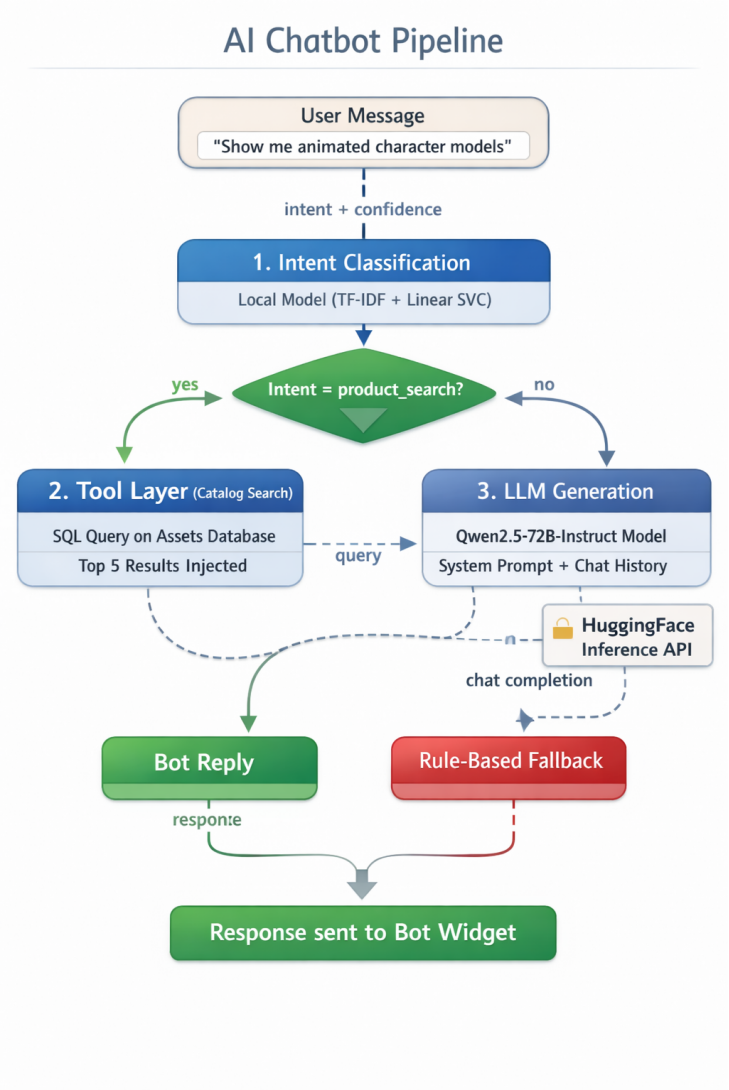

Figure 5.1: AI Chatbot Pipeline — intent classification, RAG tool layer, and LLM generation

Realization

The chatbot is available as a floating widget on every page of the marketplace, accessible to both authenticated users and guests. A submitted message passes through the three-stage pipeline and the generated answer is returned to the interface in real time.

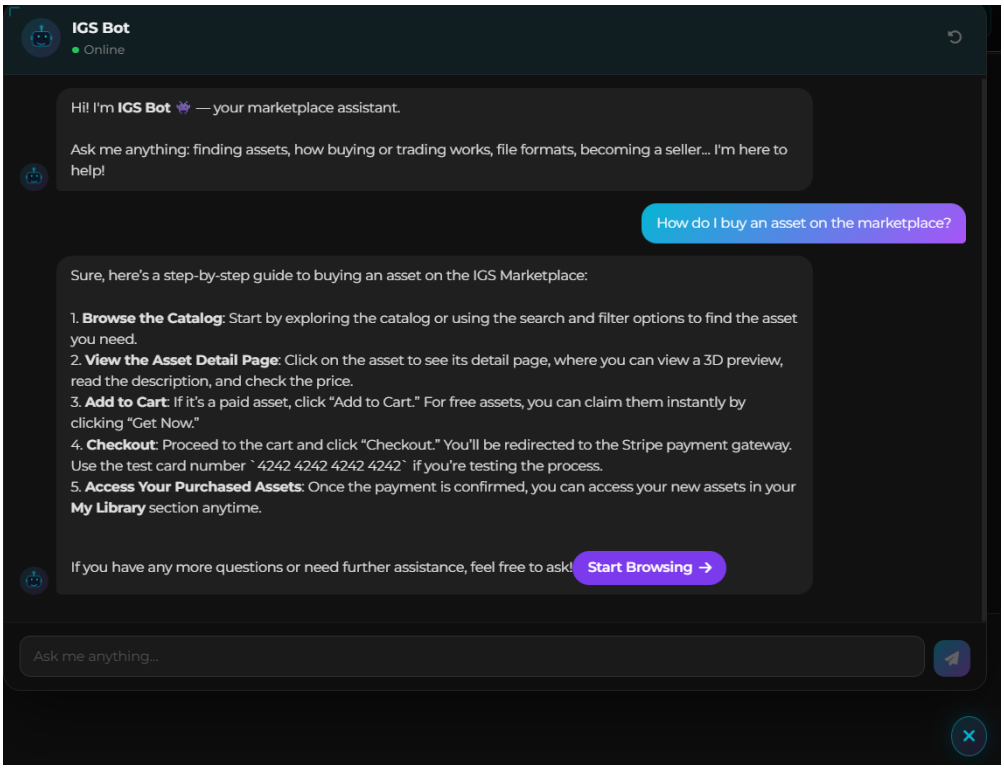

Figure 5.2: AI chatbot widget — intent-aware assistant available on every page

Conclusion

The combination of a local intent classifier and a cloud language model achieves both responsiveness and generative quality. Straightforward intents are handled immediately, while complex natural-language queries benefit from the reasoning capacity of the large model.

The second AI service classifies user reviews automatically to support moderators and sellers in interpreting large volumes of feedback.

5.1.2 Review Sentiment Analysis

Analysis

Overview. Every review submitted by a buyer is automatically analyzed to determine whether its content is positive, neutral or negative. This sentiment classification supports moderators in auditing large volumes of user feedback and provides sellers with an aggregated view of how their assets are perceived, without requiring manual reading of each entry.

Primary Model — DistilBERT. The primary classification engine is **DistilBERT** [27], a compressed variant of the **BERT** architecture.

BERT — Bidirectional Encoder Representations from Transformers

BERT [28] is a pre-trained deep bidirectional Transformer that reads the entire input sequence simultaneously, capturing context from both sides of each word. This bidirectional attention mechanism makes it significantly more effective at understanding negation and nuance than models that process text strictly left to right.

Knowledge Distillation and DistilBERT

DistilBERT [27] is obtained through **knowledge distillation**: a smaller *student* model is trained to replicate the output distribution of a larger *teacher* model (BERT), rather than learning from raw labels alone. The result retains approximately 97 % of BERT's performance on downstream tasks while using 40 % fewer parameters and running 60 % faster.

The variant used in this project was fine-tuned on the **Stanford Sentiment Treebank** (SST-2) [29], a benchmark dataset of sentence-level sentiment annotations drawn from movie reviews. Its classification of short evaluative sentences transfers naturally to marketplace reviews. The model returns a binary label (positive or negative) alongside a confidence score; reviews with a confidence below 0.65 are mapped to the *neutral* class to avoid committing to a label when the signal is ambiguous.

Fallback Model — Logistic Regression. A secondary **Logistic Regression** [23] classifier trained on TF-IDF features serves as an offline fallback when the cloud inference API is unavailable. Logistic Regression estimates the probability of class membership by applying the logistic function to a linear combination of input features. It produces a three-class output (positive, neutral, negative) and is well suited to this role because of its low computational footprint and interpretable decision boundaries. Table 5.2 summarizes the runtime decision logic.

Table 5.2: Sentiment classification — decision logic

Condition	Model	Output label
Cloud API available, confidence ≥ 0.65	DistilBERT	positive or negative
Cloud API available, confidence < 0.65	DistilBERT	neutral
Cloud API unavailable	Logistic Regression	positive, neutral or negative

Realization

Each review is analyzed immediately upon submission. The resulting sentiment label is stored alongside the review and displayed as a color-coded badge on the asset detail page. The seller dashboard aggregates all labels into a summary chart visible from the Creator Studio.

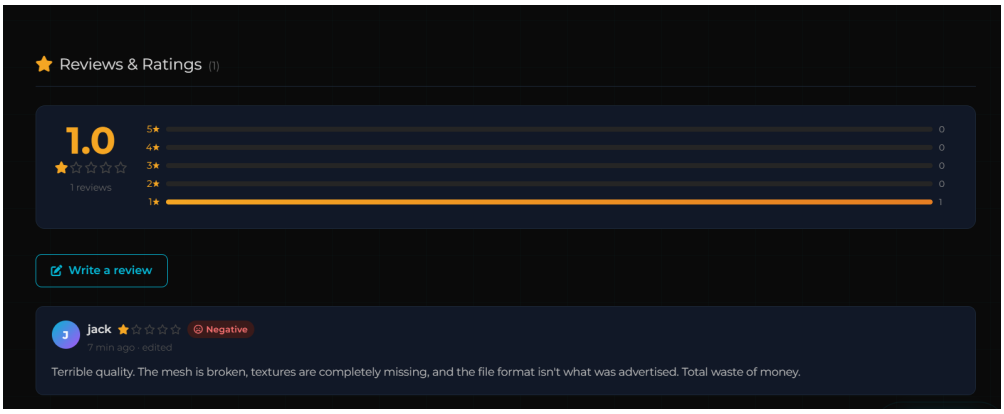

Figure 5.3: Sentiment analysis — color-coded sentiment badges displayed on the asset detail page

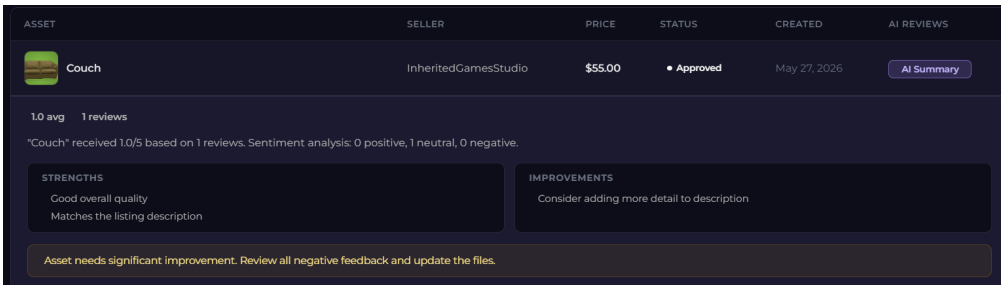

Figure 5.4: Sentiment analysis — aggregate sentiment summary visible in the Workspace admin view

Conclusion

Automated sentiment classification reduces the effort required to audit large volumes of reviews. Moderators can quickly identify trends and sellers receive actionable feedback without reading every individual entry.

The third AI service surfaces relevant assets to each user based on their interaction history.

5.1.3 Personalized Recommendations

Analysis

Overview. The recommendation system surfaces relevant assets to each user based on their interaction history. It follows a **content-based filtering** approach [30], which

derives user preferences from past behaviour and matches them against asset attributes rather than comparing users with one another. This design does not require a large user base to produce meaningful results, making it well-suited to a growing marketplace.

Similarity Measure — Jaccard Index. Each asset is represented by its set of descriptive tags.

Jaccard Similarity Index

The **Jaccard similarity index** [31] quantifies the overlap between two sets A and B as the ratio of their intersection to their union:

$$J(A,\, B) \;=\; \frac{|A \cap B|}{|A \cup B|}$$

A value of 1 indicates perfectly identical sets; a value of 0 indicates no overlap at all. The Jaccard index is well suited to recommendation tasks based on tags because the attributes are binary (present or absent) and the sets vary widely in size across categories.

User Profile Construction. The user's preference profile is built from four interaction types, each weighted according to the strength of preference it expresses:

Table 5.3: User interaction signals and their weights

Interaction	Weight	Rationale
Asset like	5	Strongest explicit preference signal
Paid purchase	4	High-confidence revealed preference
Asset download	3	Direct acquisition of content
Star rating	1–5	Weighted proportionally to the numeric value

Diversity and Cold-Start Handling. To prevent the ranked list from being dominated by near-identical assets, a post-processing diversity pass demotes any item whose tag overlap with a higher-ranked result exceeds 70 %. Users with no recorded interactions — guests and newly registered accounts — receive a trending list ranked by recent download and purchase frequency, and the interface displays "Trending Now" rather than "Recommended for You" in this cold-start state.

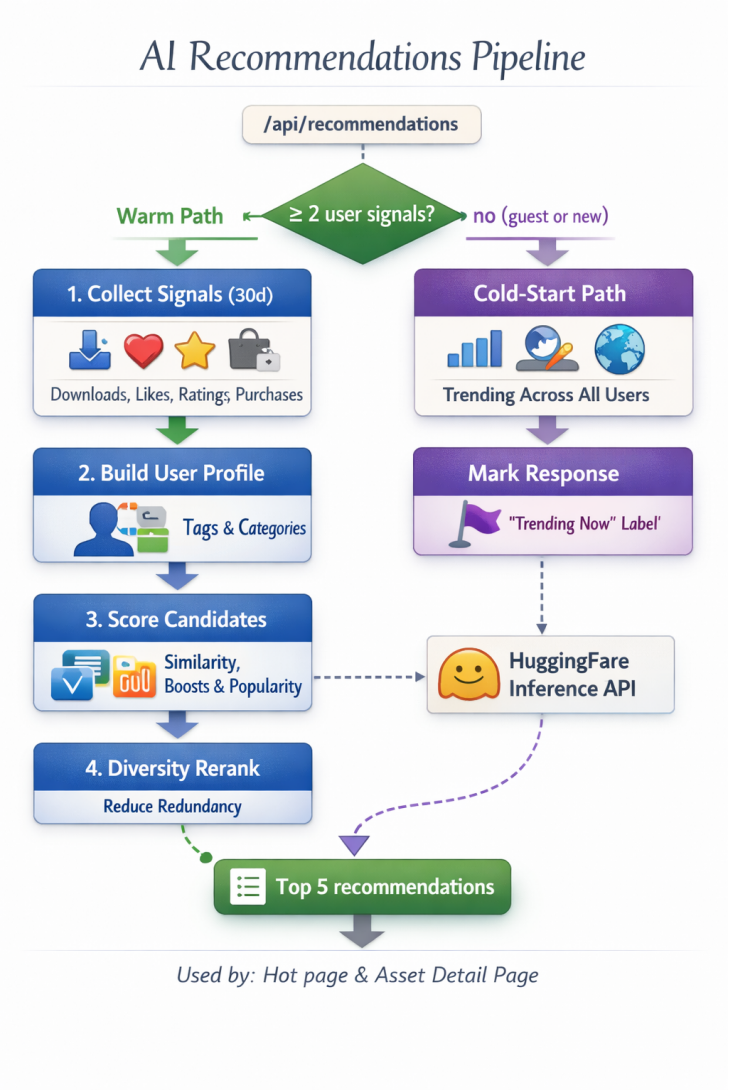

Figure 5.5: AI Recommendations Pipeline — warm path (personalized) and cold-start path (trending)

Realization

The recommendation page is accessible to both authenticated and anonymous users. Users with sufficient interaction history receive a personalized ranking; all others see the trending catalog. The interface is visually consistent with the standard catalog pages.

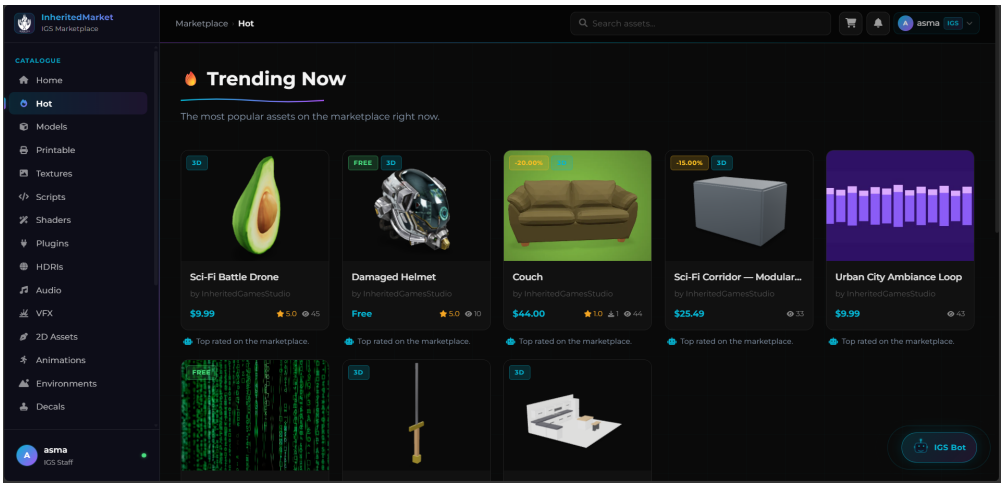

Figure 5.6: Recommendations page — personalized (warm user) and trending (cold start)

Conclusion

Content-based filtering with Jaccard similarity delivers personalized results without requiring collaborative data or model retraining. The cold-start strategy ensures every visitor receives meaningful content from the first interaction.

Alongside the AI services, three independent security mechanisms were implemented to protect the platform against common attack vectors.

5.2 Security Hardening

Three independent security mechanisms protect the platform against the most common attack vectors catalogued in the OWASP Top Ten [32].

- **Rate limiting.** Each sensitive endpoint is protected by a per-IP request ceiling. Authentication attempts are limited to ten requests per fifteen-minute window, and account registration is capped at five per hour. This mitigates brute-force and credential-stuffing attacks by making exhaustive enumeration computationally impractical.

- **Token revocation.** Authentication tokens are immediately invalidated upon logout through a server-side blacklist consulted on every authenticated request. This ensures that a token intercepted after the user has signed out cannot be reused, regardless of its expiry date.

- **Magic-byte file validation.** Uploaded images are validated at the binary level against the known **magic bytes** of accepted formats (PNG, JPEG, WebP, GIF). Magic bytes are fixed byte sequences at the start of a file that identify its true

format independently of the filename or MIME type declared by the client. A file whose binary header does not match its declared type is rejected before reaching storage, preventing MIME-confusion attacks in which a malicious file is disguised as a safe image.

Beyond the mechanisms listed above, the OWASP coverage also includes: broken access control addressed through the RBAC permission model, injection risks eliminated through parameterized database queries throughout the data layer, and security misconfiguration mitigated by security-oriented HTTP headers and mandatory environment-variable validation at application startup.

Beyond security and AI capabilities, the platform offers an immersive preview experience enabled by the WebXR standard.

5.3 VR/AR Immersive Preview

This section examines the technical foundations of the immersive preview and presents the validated implementation.

5.3.1 Analysis

Browser-Native Immersive Web. The immersive preview feature is delivered entirely through the browser, with no plugin or companion application required on the user's side. This is made possible by a recent W3C standard that exposes virtual and augmented reality hardware directly to web pages.

WebXR Device API

The **WebXR Device API** [33] is a W3C standard that provides browsers with a unified interface for accessing virtual and augmented reality hardware. It abstracts the differences between headset models, controller layouts and spatial tracking systems, allowing a single web application to deliver immersive experiences across a variety of XR devices without platform-specific adaptations.

Because the API is implemented natively in modern browsers, the immersive preview is accessible to anyone owning a compatible headset, without any installation step.

3D Rendering and Stereo Mode.

WebGL — Web Graphics Library

WebGL [34] is a JavaScript API standardized by the Khronos Group that provides direct access to the device's GPU from within a web browser, without plugins. It exposes a subset of the OpenGL ES 2.0 API, enabling the browser to execute shader programs on the GPU and render complex 3D scenes at interactive frame rates. Because WebGL runs inside the browser's security sandbox, it achieves near-native rendering performance while remaining safe for untrusted web content.

The asset viewer is powered by **Three.js** [35], a JavaScript library that abstracts the low-level WebGL rendering pipeline. When a user enters immersive mode, the renderer transitions to **stereoscopic projection**, producing a separate image frame for each eye at the correct inter-pupillary offset to create a convincing perception of depth. In augmented reality mode, the rendered 3D asset is composited over the live camera feed provided by the headset's passthrough cameras, allowing the user to inspect the object within their physical environment.

To maintain rendering correctness in XR sessions, visual post-processing effects that rely on multi-pass framebuffer operations are deactivated during immersive rendering. These effects are not designed for the stereo framebuffer required by headsets and would otherwise cause visual artefacts or rendering errors.

Hardware Validation. The feature was validated on a **Meta Quest 3** [8] mixed-reality headset. The device was connected to the same local network as the development server and accessed the application through its built-in browser over HTTPS, which is required by the WebXR specification. Two modes were tested: a fully immersive **VR mode** in which the user is surrounded by the rendered 3D scene, and a **passthrough AR mode** in which the asset is overlaid on the real environment captured by the device's colour cameras. Both modes operated correctly, confirming that the platform delivers an authentic WebXR experience on consumer mixed-reality hardware.

The following subsection demonstrates the immersive preview experience validated on consumer mixed-reality hardware.

5.3.2 Realization

On any compatible 3D asset detail page, the asset is rendered interactively in the browser using a Three.js scene with orbit controls, post-processing effects, and environment lighting. The viewer supports `.glb` and `.gltf` formats and allows buyers to examine geometry, materials, and surface detail at any angle before purchasing.

Figure 5.7: 3D model viewer — interactive orbit controls and environment lighting in the browser

From the same page, the user can enter VR or AR mode directly from the browser with a single click. In VR mode, the asset is rendered at true real-world scale inside the headset, allowing buyers to evaluate proportions and surface detail. In AR mode, the asset is placed within the user's physical environment using the headset's passthrough cameras.

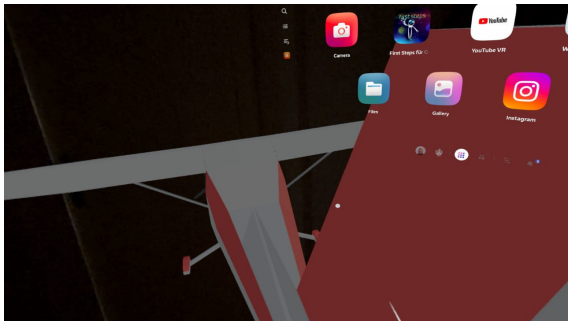

Figure 5.8: 3D asset rendered at full scale on Meta Quest 3 - 1

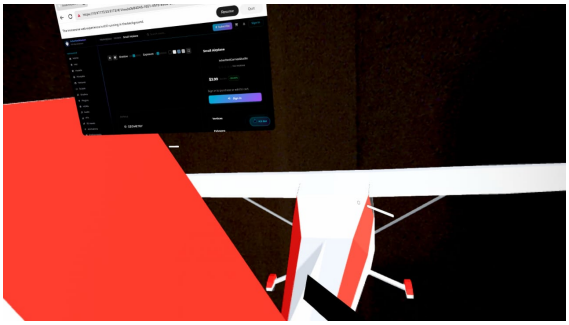

Figure 5.9: 3D asset rendered at full scale on Meta Quest 3 - 2

Once all features were validated, the platform was deployed to a production environment, as described in the following section.

5.4 Deployment on Hostinger VPS

This section describes the production infrastructure, server configuration and the automated delivery pipeline.

5.4.1 Infrastructure Overview

The production environment is organized around three independent tiers, each handling a distinct concern. The compute tier consists of a **Hostinger Virtual Private Server** running Ubuntu 22.04 LTS, which hosts the Node.js backend and serves the two compiled React applications as static files. The data tier relies on **Supabase**, a cloud-hosted service that manages both the PostgreSQL database and the private file storage buckets, eliminating the need to maintain a database instance on the server itself. Finally, the AI tier delegates all inference workloads to the **Hugging Face Inference API**, meaning no GPU hardware is required on the VPS.

The server configuration details how incoming traffic is handled, how HTTPS is enforced and how the application process is kept alive.

5.4.2 Server Configuration

Reverse Proxy and HTTPS. **Nginx** acts as the single entry point for all incoming traffic. It terminates HTTPS connections using a free TLS certificate issued by **Let's Encrypt** and renewed automatically every ninety days. Secure connections are not optional: the WebXR specification mandates HTTPS before a browser will grant access to XR hardware. Nginx routes API and real-time traffic to the backend process while serving the frontend builds directly from disk.

PM2 — Node.js Process Manager

PM2 is a production-grade process manager for Node.js applications. It keeps the server process alive after crashes, restarts it automatically on reboot, and enables zero-downtime reloads so that new deployments do not interrupt active user sessions. Log rotation and resource monitoring are handled natively.

Secrets Management. All sensitive configuration — database credentials, JWT signing keys, payment gateway tokens, and third-party API keys — is stored in environment variables on the server and never committed to the source repository. This ensures that the Git history remains free of credentials and that different environments (development, production) can carry their own configuration without code changes.

The final element of the deployment setup is the automated continuous integration and delivery pipeline.

5.4.3 CI/CD Pipeline

Continuous Integration (CI) is the practice of automatically building and testing every code change as soon as it is pushed to the shared repository. The goal is to

detect integration problems immediately, before they accumulate into larger conflicts. **Continuous Delivery (CD)** extends CI by automatically preparing and deploying a validated build to the target environment, so that every passing build can be released to users without manual intervention.

Automated Delivery with GitHub Actions. The project uses **GitHub** for version control and **GitHub Actions** as the CI/CD platform. A workflow triggered on every merge to the main branch performs three sequential stages: the *CI stage* installs dependencies and executes the integration test suite to confirm the build is healthy; the *packaging stage* compiles both React frontends into optimized static bundles; and the *CD stage* connects to the VPS over SSH to pull the latest revision, deploy the new bundles, and reload the backend process with zero downtime. Deployment credentials are stored as encrypted repository secrets and are never exposed in workflow logs.

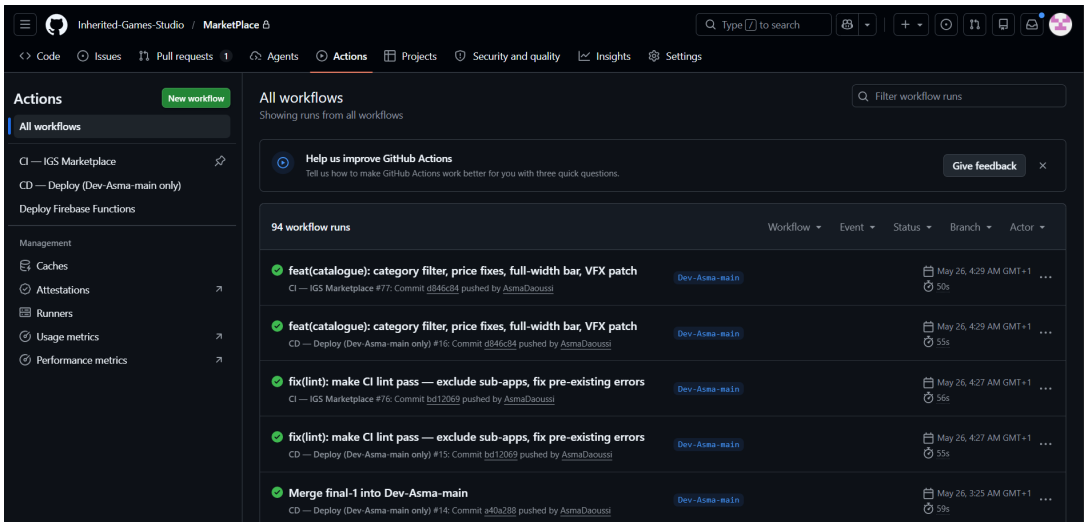

Figure 5.10: GitHub Actions CI/CD pipeline — automated build, test, and deployment workflow

Conclusion

This chapter completed the platform's feature set by delivering the AI services, immersive previews, and production infrastructure that distinguish the IGS Marketplace from conventional asset stores. The AI pipeline — combining intent classification, sentiment analysis, and content-based recommendations — adds intelligence to the user experience without requiring external collaborative data. With the VPS deployment and CI/CD pipeline in place, the platform is ready for production. A general conclusion summarizing the project's achievements and future perspectives follows.

General Conclusion

IGS Marketplace was successfully designed and developed over five sprints, delivering Tunisia's first dedicated 3D digital asset platform. The system provides a six-level Role-Based Access Control system, an interactive in-browser 3D asset viewer, Stripe payment integration, AI-powered review sentiment analysis, a personalized recommendation engine, a floating chatbot assistant powered by Qwen2.5-72B-Instruct, and a companion internal workspace application — all within a cohesive full-stack ecosystem built on React, Node.js/Express, and PostgreSQL.

Several key technical insights emerged throughout the project. Designing the permission system during Sprint 1 and enforcing it consistently across all subsequent sprints was the single most impactful architectural decision — retrofitting access control would have been significantly more costly. The three-tier layered architecture (Routes → Controllers → Repositories) proved highly maintainable for a REST API at this level of complexity, and applying Scrum as a solo developer, with supervisors occupying the Product Owner and Scrum Master roles, effectively structured six months of work into reviewable, deliverable-focused increments.

Looking ahead, several directions are planned for the platform's continued evolution. The internal workspace application will grow into a full **ERP system** with richer project management, HR, and operational modules, while its customer-facing side will expand into a comprehensive **CRM** supporting advanced customer segmentation, relationship tracking, and communication workflows. On the payment front, **Flouci** integration for Tunisian dinar transactions is already partially in place and will be fully activated once the required administrative and regulatory procedures are completed. The asset catalog will be broadened with new content categories to serve a wider range of creative disciplines. A **bundle system** will allow sellers to group related assets and offer them at a combined price, increasing average order value and discoverability. Finally, the **voucher system** — currently scoped to internal users — will be extended to all user types, enabling promotional campaigns for members, sellers, and subscribers alike. These evolutions position the IGS Marketplace as a long-term, scalable platform well beyond its current scope.

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Abstract

This end-of-studies project presents the design and development of **IGS Marketplace**, a full-stack digital platform built for **Inherited Games Studio (IGS)**, a Tunisian video-game company. The solution addresses the absence of a national marketplace for 3D creative content by offering local creators a dedicated space to publish and monetize their work, with secure payments in both international and Tunisian currency.

The platform consists of two complementary applications sharing a common back-end: a public *Marketplace* covering thirteen asset categories with interactive 3D and WebXR previews on virtual-reality headsets, and an internal *Workspace* for project management, content moderation and administration. Three AI services are integrated — a conversational chatbot, a DistilBERT-based review sentiment analyzer and a Jaccard-based recommendation engine. The technical stack relies on **React**, **Node.js / Express** and **PostgreSQL (Supabase)**, following the **Scrum** methodology over five sprints.

Keywords: 3D Marketplace, WebXR, React, Node.js, PostgreSQL, RBAC, Scrum, Artificial Intelligence.

Résumé

Ce projet de fin d'études présente la conception et le développement de **IGS Marketplace**, une plateforme web full-stack réalisée pour **Inherited Games Studio (IGS)**, société tunisienne de développement de jeux vidéo. La solution répond à l'absence d'une marketplace nationale dédiée aux contenus 3D, en offrant aux créateurs locaux un espace de publication et de monétisation avec un paiement sécurisé en devises internationales et en dinar.

La plateforme se compose de deux applications partageant un back-end commun: une *Marketplace* publique couvrant treize catégories d'assets avec prévisualisation 3D interactive et mode immersif WebXR sur casques de réalité virtuelle, et un *Workspace* interne pour la gestion de projet, la modération et l'administration. Trois services d'intelligence artificielle sont intégrés — un chatbot conversationnel, une analyse de sentiment basée sur DistilBERT et un moteur de recommandations reposant sur la similarité de Jaccard. La pile technique repose sur **React**, **Node.js / Express** et **PostgreSQL (Supabase)**, suivant la méthodologie **Scrum** sur cinq sprints.

Mots-clés : Marketplace 3D, WebXR, React, Node.js, PostgreSQL, RBAC, Scrum, Intelligence Artificielle.

ملخص

يقدم هذا المشروع تصميم وتطوير منصة **Marketplace IGS** لفائدة شركة **Studio Games Inherited** التونسية المتخصصة في تطوير ألعاب الفيديو. تعالج المنصة غياب فضاء وطني مخصص للأصول الإبداعية ثلاثية الأبعاد، وتوفر للمبدعين المحليين فضاءً للنشر والبيع مع دعم الدفع الآمن بالعملات الدولية وبالدينار التونسي.

تتكون المنصة من تطبيقين متكاملين يشتركان في خادم خلفي موحد: واجهة عامة *Marketplace* تغطي ثلاث عشرة فئة من الأصول الرقمية مع عرض تفاعلي ومعاينة غامرة عبر تقنية WebXR، ولوحة تحكم داخلية *Workspace* لإدارة المشاريع ومراقبة المحتوى. تدمج المنصة ثلاث خدمات للذكاء الاصطناعي: روبوت محادثة، ومحلل لمشاعر التقييمات يعتمد على نموذج DistilBERT، ومحرك توصيات مبني على مقياس Jaccard. وتعتمد التقنيات المستخدمة على **React** و**Node.js / Express** و**PostgreSQL (Supabase)**، باتباع منهجية **Scrum** عبر خمس دورات تطويرية.

الكلمات المفتاحية: منصة ثلاثية الأبعاد، WebXR، React، Node.js، PostgreSQL، RBAC، Scrum، ذكاء اصطناعي.
